S.C.O.U.T. · SYSTEM BLOCK DIAGRAM · SCOUT-S-000

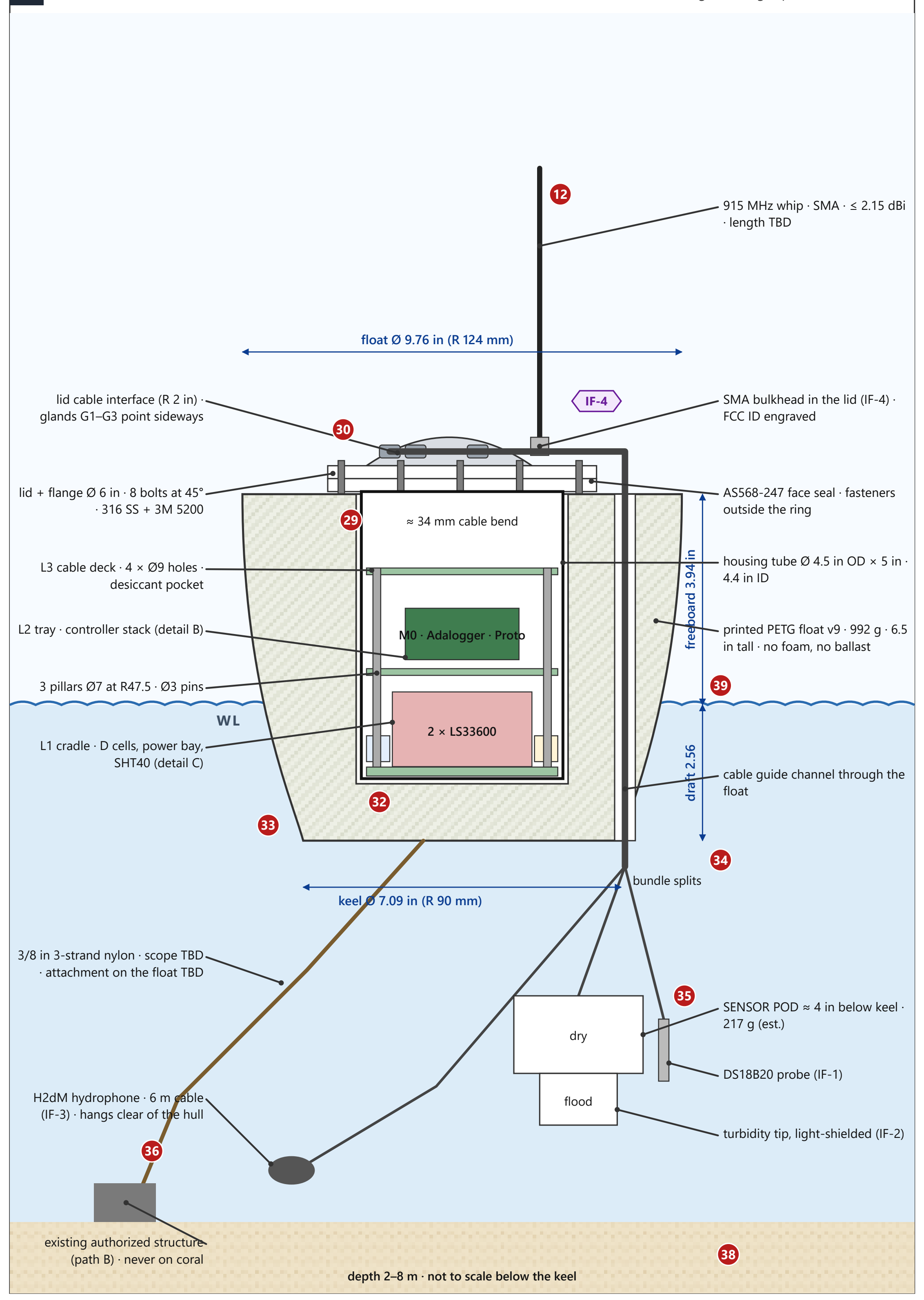
Integrated system: mechanical · electrical · firmware · data

Everything on sheets 2–4 is on this sheet, with the same letters and note numbers

PRELIMINARY · NOT FOR FINAL BUILD

M AS-DEPLOYED BUOY · SECTION

float v9 + flange housing + pod · schematic, NTS


915 MHz whip · SMA · ≤ 2.15 dBi · length TBD
float Ø 9.76 in (R 124 mm)
lid cable interface (R 2 in) · glands G1–G3 point sideways
IF-4
SMA bulkhead in the lid (IF-4) · FCC ID engraved
lid + flange Ø 6 in · 8 bolts at 45° · 316 SS + 3M 5200
AS568-247 face seal · fasteners outside the ring
≈ 34 mm cable bend
L3 cable deck · 4 × Ø9 holes · desiccant pocket
housing tube Ø 4.5 in OD × 5 in · 4.4 in ID
L2 tray · controller stack (detail B)
M0 · Adalogger · Proto
freeboard 3.94 in
printed PETG float v9 · 992 g · 6.5 in tall · no foam, no ballast
3 pillars Ø7 at R47.5 · Ø3 pins
WL
L1 cradle · D cells, power bay, SHT40 (detail C)
2 × LS33600
draft 2.56
cable guide channel through the float
bundle splits
keel Ø 7.09 in (R 90 mm)
3/8 in 3-strand nylon · scope TBD · attachment on the float TBD
SENSOR POD ≈ 4 in below keel · 217 g (est.)
dry
DS18B20 probe (IF-1)
flood
turbidity tip, light-shielded (IF-2)
H2dM hydrophone · 6 m cable (IF-3) · hangs clear of the hull
existing authorized structure (path B) · never on coral
depth 2–8 m · not to scale below the keel
12
30
29
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32
33
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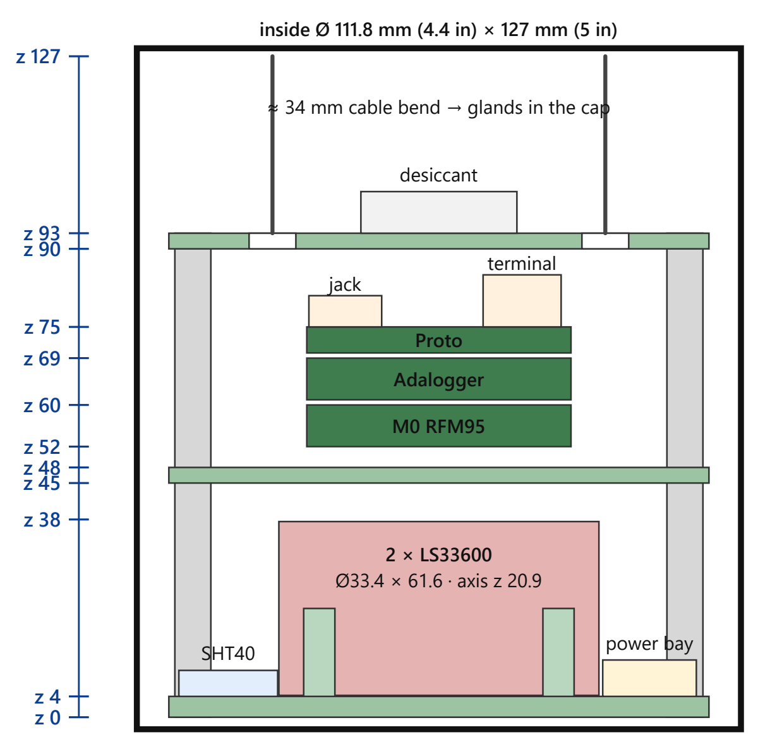
36
38

N ELECTRONICS HOUSING · PACKAGING OPTION C

side section · z above housing floor, mm



93 of 127 mm used · ≈ 34 mm left for cable bend under the cap

L3 deck z 90–93 · 3 mm · 4 × Ø9 cable holes at (±32, ±12) · desiccant pocket 30 × 14 × 8

L2 tray z 45–48 · 4 × Ø6 × 4 standoffs at (±22.9, ±8.9) · boards z 52–75 · jack and terminal on the Proto

pillars Ø7 at R47.5 (90°, 210°, 330°) · 41 mm L1–L2, 38 mm L2–L3 · Ø3 pins or M3 rod in Ø3.2 holes

L1 cradle z 0–38 · 4 mm floor · 2 saddle ribs 69 × 6 × 16.9 with Ø33.8 notches · cells along X at y ±17.2

power bay at x 40.5: diode pockets 15 × 8, PTC 9 × 4, JST 10 × 7 · **SHT40** pocket at x −40.5

(28) (32)

P SEALS · PENETRATIONS · MATERIALS

mechanical/cad READMEs + facts.md

Item	Spec	Status
Housing seal	AS568-247, ID 117.07 × CS 3.53 mm · groove ID 117.07 / OD 122.00, 4.93 wide × 2.65 deep → 25% squeeze, 75% fill · fasteners outside the ring	confirm ring (SCO-137)
Flange + lid	Ø6 in flange, 8 bolts at 45° (Ø0.22 / Ø0.13 in holes) · tube Ø4.5 in OD × 5 in · R2 in curved cable interface · FCC ID engraved	DFM drawing, not released
Pod seal	Everbilt 2-5/8 OD × 3/16 in CS · groove Ø2.20 × Ø2.74 in, ≈ 28% squeeze · 6 bolts on Ø3.10 in · leads potted in epoxy with an anti-wick boss	leaked twice; re-test with real ring
Glands	lab stock, sized per cable · 3 in the cap (+ SMA bulkhead)	positions not drawn
Hardware	316 stainless · 3M 5200 over bolt heads and joints	decided 10-16
Body + coating	PETG (ASA for scale-up) · Sea Hawk Smart Solution antifouling, copper-free	decided
Pressure	5 m water-equivalent (50.3 kPa) before any deployment	not tested

(37)

Q MASS · BUOYANCY · STABILITY

printed-float tip explorer, v9 re-run 2026-10-16

Quantity	Value
Total mass (estimate)	2.87 kg
Printed float v9	992 g
Draft / freeboard	2.56 / 3.94 in
KG / GM	2.26 / 1.47 in
Max righting arm	40 mm at ≈ 70°
GZ at 178° (v8)	≈ 1 mm
Float envelope	7.0 L · 6.5 in tall
+2 kg payload	2.1 in freeboard left

Mass inputs (estimates): battery 250 g at 25 mm · boards at 80–90 mm · antenna at 150 mm · epoxy/coating 330 g · mooring hardware 220 g at the keel · pod 217 g + hydrophone 150 g, 4 in below keel. (33)

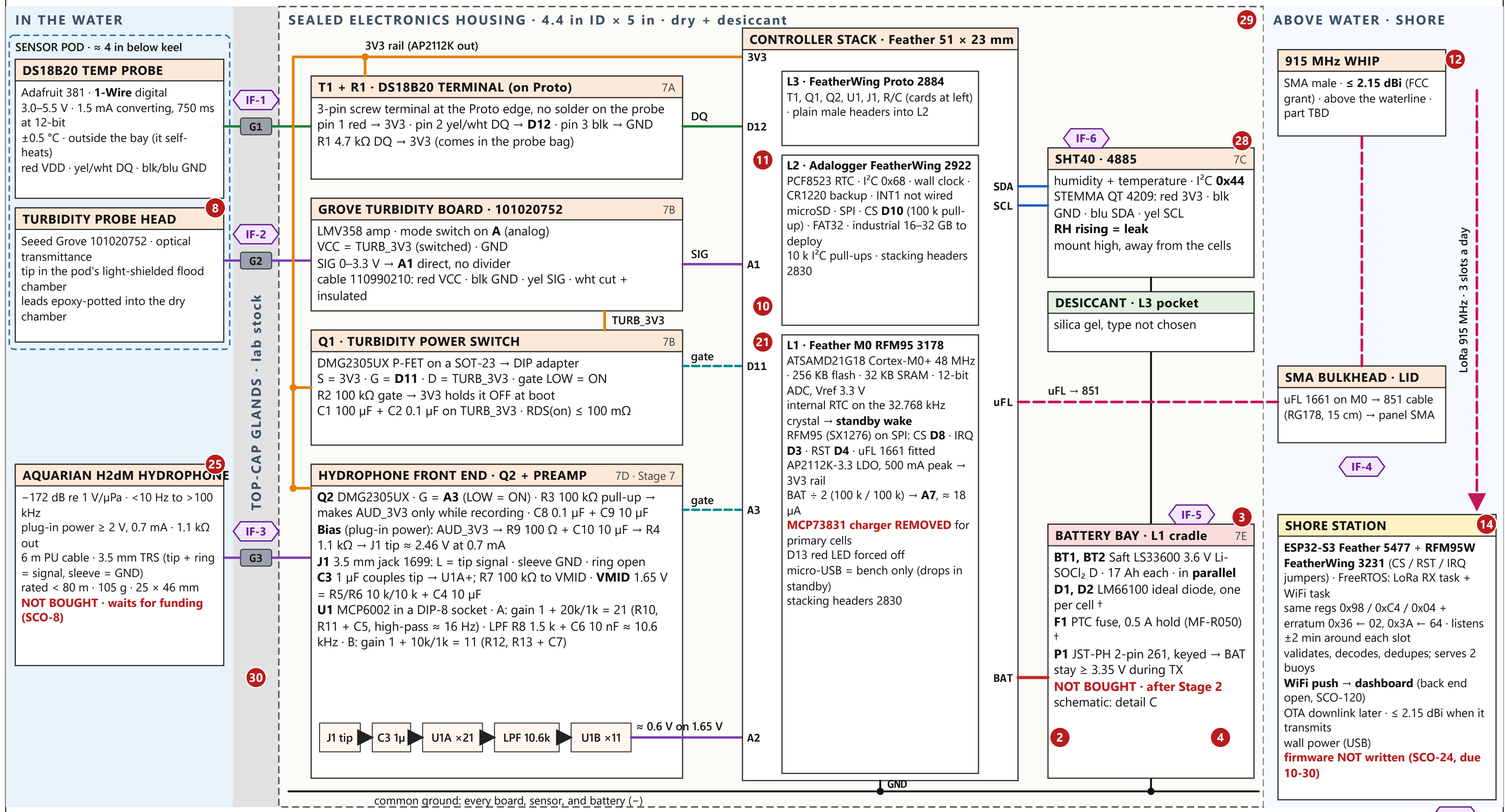
No ballast, no lead, no foam. 500 g at the keel only lifts GZ(178°) from 1.1 to 1.5 mm, so hull shape governs. Pod depth moves GM: at the keel 1.20 in, 13.5 in below 1.96 in.

Not covered: mooring heel, wave loads, chamber flooding, fouling mass. Physical tilt test from 180° (SCO-171).

(34)

A ELECTRICAL BLOCK DIAGRAM

every part, pin, net and protocol, buoy to shore radio



SENSOR POD · ≈ 4 in below keel

DS18B20 TEMP PROBE
Adafruit 381 · **1-Wire** digital
3.0–5.5 V · 1.5 mA converting, 750 ms at 12-bit
±0.5 °C · outside the bay (it self-heats)
red VDD · yel/wht DQ · blk/blu GND

TURBIDITY PROBE HEAD
Seeed Grove 101020752 · optical transmittance
tip in the pod's light-shielded flood chamber
leads epoxy-potted into the dry chamber

AQUARIAN H2dM HYDROPHONE
−172 dB re 1 V/µPa · <10 Hz to >100 kHz
plug-in power ≥ 2 V, 0.7 mA · 1.1 kΩ out
6 m PU cable · 3.5 mm TRS (tip + ring = signal, sleeve = GND)
rated < 80 m · 105 g · 25 × 46 mm
NOT BOUGHT · waits for funding (SCO-8)

T1 + R1 · DS18B20 TERMINAL (on Proto) 7A
3-pin screw terminal at the Proto edge, no solder on the probe
pin 1 red → 3V3 · pin 2 yel/wht DQ → **D12** · pin 3 blk → GND
R1 4.7 kΩ DQ → 3V3 (comes in the probe bag)

GROVE TURBIDITY BOARD · 101020752 7B
LMV358 amp · mode switch on **A** (analog)
VCC = TURB_3V3 (switched) · GND
SIG 0–3.3 V → **A1** direct, no divider
cable 110990210: red VCC · blk GND · yel SIG · wht cut + insulated

Q1 · TURBIDITY POWER SWITCH 7B
DMG2305UX P-FET on a SOT-23 → DIP adapter
S = 3V3 · G = **D11** · D = TURB_3V3 · gate LOW = ON
R2 100 kΩ gate → 3V3 holds it OFF at boot
C1 100 µF + C2 0.1 µF on TURB_3V3 · RDS(on) ≤ 100 mΩ

HYDROPHONE FRONT END · Q2 + PREAMP 7D · Stage 7
Q2 DMG2305UX · G = **A3** (LOW = ON) · R3 100 kΩ pull-up → makes AUD_3V3 only while recording · C8 0.1 µF + C9 10 µF
Bias (plug-in power): AUD_3V3 → R9 100 Ω + C10 10 µF → R4 1.1 kΩ → J1 tip ≈ 2.46 V at 0.7 mA
J1 3.5 mm jack 1699: L = tip signal · sleeve GND · ring open
C3 1 µF couples tip → U1A+; R7 100 kΩ to VMID · **VMID** 1.65 V = R5/R6 10 k/10 k + C4 10 µF
U1 MCP6002 in a DIP-8 socket · A: gain 1 + 20k/1k = 21 (R10, R11 + C5, high-pass ≈ 16 Hz) · LPF R8 1.5 k + C6 10 nF ≈ 10.6 kHz · B: gain 1 + 10k/1k = 11 (R12, R13 + C7)

CONTROLLER STACK · Feather 51 × 23 mm

L3 · FeatherWing Proto 2884
T1, Q1, Q2, U1, J1, R/C (cards at left) · plain male headers into L2

L2 · Adalogger FeatherWing 2922
PCF8523 RTC · I^{2}C 0x68 · wall clock · CR1220 backup · INT1 not wired
microSD · SPI · CS **D10** (100 k pull-up) · FAT32 · industrial 16–32 GB to deploy
10 k I^{2}C pull-ups · stacking headers 2830

L1 · Feather M0 RFM95 3178
ATSAMD21G18 Cortex-M0+ 48 MHz · 256 KB flash · 32 KB SRAM · 12-bit ADC, Vref 3.3 V
internal RTC on the 32.768 kHz crystal → **standby wake**
RFM95 (SX1276) on SPI: CS **D8** · IRQ **D3** · RST **D4** · uFL 1661 fitted
AP2112K-3.3 LDO, 500 mA peak → 3V3 rail
BAT ÷ 2 (100 k / 100 k) → **A7**, ≈ 18 µA
MCP73831 charger REMOVED for primary cells
D13 red LED forced off
micro-USB = bench only (drops in standby)
stacking headers 2830

SHT40 · 4885 7C
humidity + temperature · I^{2}C **0x44**
STEMMA QT 4209: red 3V3 · blk GND · blu SDA · yel SCL
RH rising = leak
mount high, away from the cells

DESICCANT · L3 pocket
silica gel, type not chosen

BATTERY BAY · L1 cradle 7E
BT1, BT2 Saft LS33600 3.6 V Li-$SOCl_2$ D · 17 Ah each · in **parallel**
D1, D2 LM66100 ideal diode, one per cell †
F1 PTC fuse, 0.5 A hold (MF-R050) †
P1 JST-PH 2-pin 261, keyed → BAT
stay ≥ 3.35 V during TX
NOT BOUGHT · after Stage 2
schematic: detail C

915 MHz WHIP
SMA male · **≤ 2.15 dBi** (FCC grant) · above the waterline · part TBD

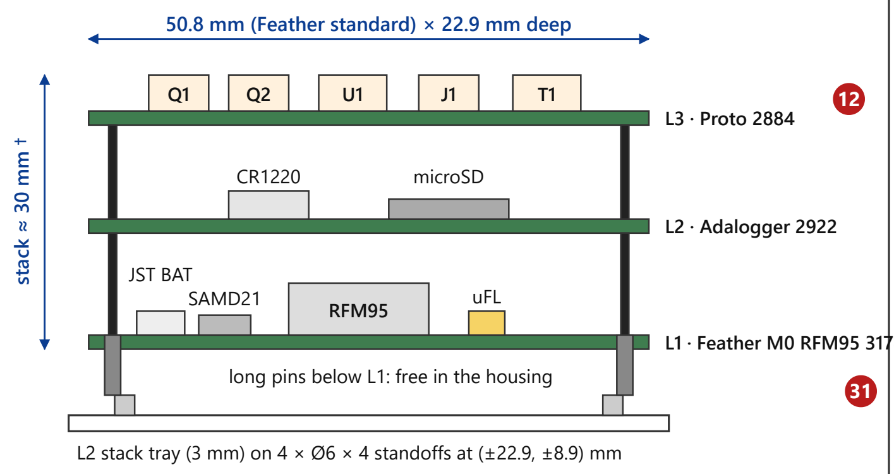
SMA BULKHEAD · LID
uFL 1661 on M0 → 851 cable (RG178, 15 cm) → panel SMA

SHORE STATION
ESP32-S3 Feather 5477 + RFM95W FeatherWing 3231 (CS / RST / IRQ jumpers) · FreeRTOS: LoRa RX task + WiFi task
same regs 0x98 / 0xC4 / 0x04 + erratum 0x36 ← 02, 0x3A ← 64 · listens ±2 min around each slot
validates, decodes, dedupes; serves 2 buoys
WiFi push → dashboard (back end open, SCO-120)
OTA downlink later · ≤ 2.15 dBi when it transmits
wall power (USB)
firmware NOT written (SCO-24, due 10-30)

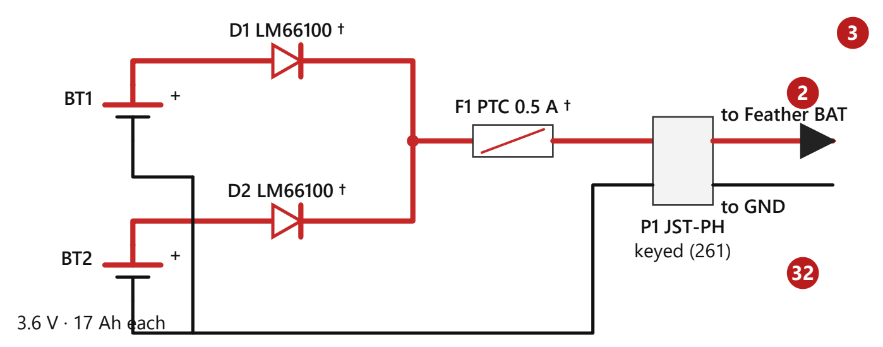
Inside the stack, no wires to add: SPI SCK / MOSI / MISO is shared by the radio (CS D8) and the SD card (CS D10); I^{2}C SDA / SCL is shared by the PCF8523 (0x68) and the SHT40 (0x44).
Pins used by the boards: D3, D4, D8 (radio), D9 / A7 (battery), D10 (SD), D13 (LED, off). Free: A0, A4, A5, D5 (old RTC wake), D6 (keep for radio IO1), D0 / D1 (Serial1).

B CONTROLLER STACK · SIDE VIEW

Heights above the housing floor (option C): M0 z 52–60 · Adalogger 61–69 · Proto 70–75 mm.
Antenna: bench = spring 4269 on the ANT pad; deployed = uFL 1661 → 851 → SMA bulkhead → whip. Use one, never both.


50.8 mm (Feather standard) × 22.9 mm deep
stack ≈ 30 mm †
Q1
Q2
U1
J1
T1
L3 · Proto 2884
CR1220
microSD
L2 · Adalogger 2922
JST BAT
SAMD21
RFM95
uFL
L1 · Feather M0 RFM95 3178
long pins below L1: free in the housing
L2 stack tray (3 mm) on 4 × Ø6 × 4 standoffs at (±22.9, ±8.9) mm
12
31

C BATTERY PACK · SCHEMATIC



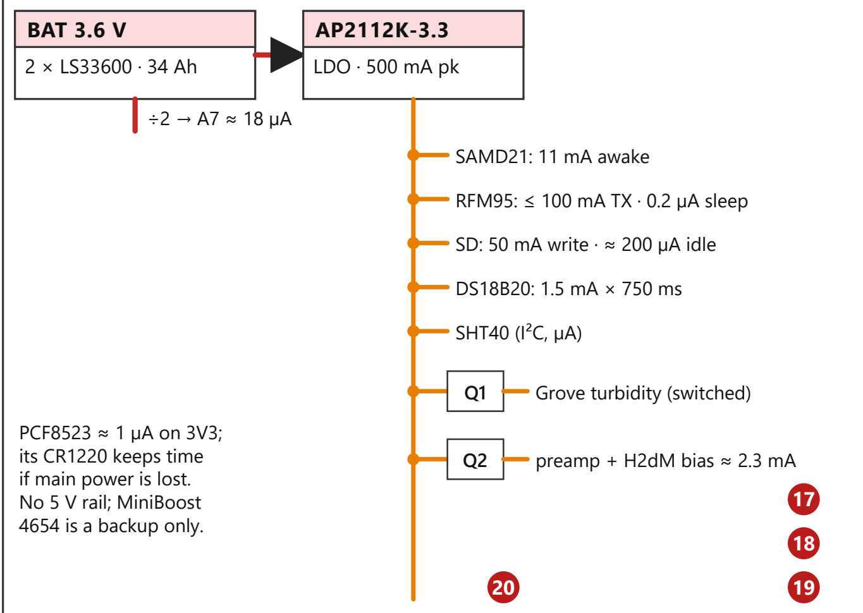
Parallel, never series (7.2 V is too high). Each ideal diode stops one cell charging the other; F1 limits a short; P1 cannot go in backwards. Measure P1 before plugging in: +3.5 to 3.7 V. Solder to the tabs only, never the cell body.

X SYSTEM AT A GLANCE

Mission	nearshore reef monitoring buoy: water temperature, turbidity, reef sound
Sites	Hawaii for reef data (Kāne'ohe Yacht Club slip first) · Monterey Bay / Santa Cruz for survivability tests · 2–8 m
Cadence	sample every 30 min · 3 uplink slots a day · audio 3 × a day (planned)
Link	LoRa 915 MHz, BW500 SF12 CR4/8, +11 dBm · ≈ 2 km LOS (datasheet) · 30 B packet
Power	2 × D Li-$SOCl_2$, no solar · ≈ 0.56 mA average without audio · 12-month target
Body	printed PETG float, ≈ 2.9 kg, self-righting · sealed 4.4 × 5 in housing
Shore	ESP32-S3 shore station on wall power → WiFi push → dashboard (back end open) · 2 buoys planned
Dates	MVP 2026-12-04 · Hawaii deployment Mar–May 2027
Cost	prototype electronics ≈ $502–652 · system target < $5,000

D POWER TREE + DAILY ENERGY BUDGET

power budget update 2026-10-05



PCF8523 ≈ 1 µA on 3V3; its CR1220 keeps time if main power is lost. No 5 V rail; MiniBoost 4654 is a backup only.

Load	mA	s	×/d	mAh/d
SAMD21 awake	11	1.5	48	0.220
DS18B20	1.5	0.75	48	0.015
Turbidity*	71.3	1.0	48	0.951
SD write	50	0.1	48	0.067
RFM95 TX	≤100	0.559	3	0.047
MCU during TX	11	0.559	3	0.005
Sleep floor	0.501	24 h	1	12.02
Total, no audio	**avg ≈ 0.56 mA**			**13.33**
Hydrophone	≈75.7	13.4 min/d		16.9
Total, with audio	**best estimate**			**30.3**

*Budget row is the SEN0189 through a 5 V boost (worst case); the Grove at 3.3 V draws less. Audio row assumed a PCM1808 ADC; the MCP6002 path is lower. Usable 23.8 Ah → ≈ 4.9 yr without audio, ≈ 2.1 yr with.

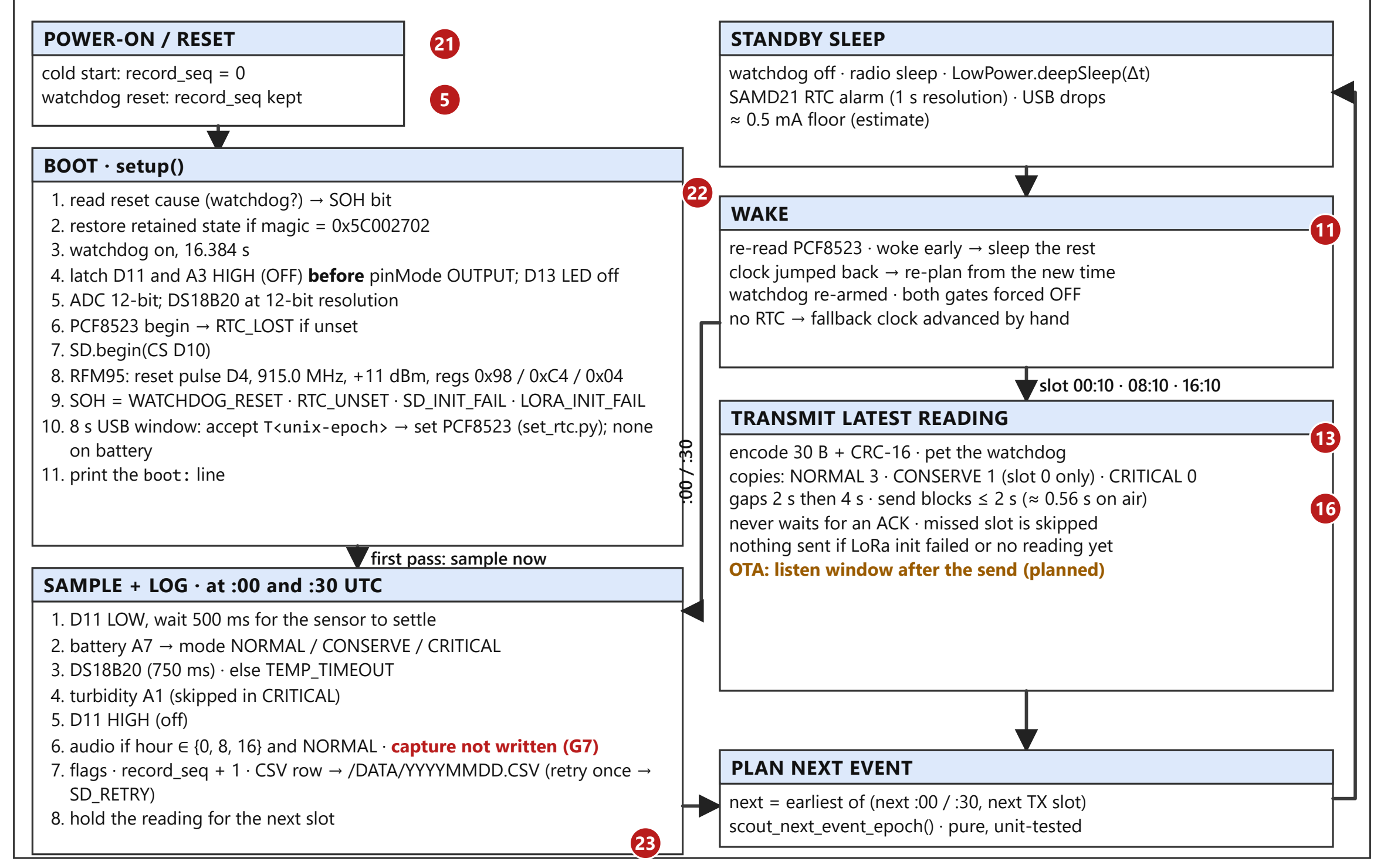
S INTERFACE REGISTER · HOW THE THREE VIEWS CONNECT

each IF tag appears on the mechanical, electrical and firmware drawings

IF	Mechanical (where it physically is)	Electrical (part → pin)	Firmware / data (what it becomes)	Status
IF-1	DS18B20 probe at the pod · cable up the float channel · gland G1 in the cap	T1 screw terminal → **D12** (1-Wire), R1 4.7 kΩ	temp_c (CSV) · temp_c_centi (packet) · TEMP_TIMEOUT	parts in hand
IF-2	Turbidity tip in the pod flood chamber · potted leads · gland G2	Grove board → **A1**; power via Q1 (**D11** LOW)	turbidity_adc (packet) · turbidity_v (SD) · NTU later	board location open
IF-3	H2dM on 6 m cable, below the pod · gland G3	J1 jack → preamp → **A2**; power via Q2 (**A3** LOW)	/AUDIO/YYYYMMDD/HHMMSSZ.WAV on SD · audio_present flag · never over LoRa	not bought · no code
IF-4	SMA bulkhead in the lid · whip above the waterline	RFM95 uFL → 851 → SMA, ≤ 2.15 dBi, +11 dBm	30 B packet · 3 slots × 3 copies · OTA downlink planned	regs unverified
IF-5	2 D cells on their side in the L1 cradle, vents away	diodes → PTC → JST → **BAT**; ÷2 → **A7**	battery_mv → power mode NORMAL / CONSERVE / CRITICAL	cells not bought
IF-6	SHT40 in the L1 floor pocket, sealed bay	STEMMA QT → I^{2}C **0x44**	internal_humidity_pct (future column) → leak alarm	no firmware
IF-7	microSD in the Adalogger, reached by opening the lid	SPI, CS **D10**	/DATA/YYYYMMDD.CSV, the complete record → sd_import on retrieval	fixed in code (G4)
IF-8	Shore station antenna at the shore site (wall power)	ESP32-S3 + RFM95W, erratum applied	decode + dedupe → WiFi push → dashboard (back end open)	no S3 firmware

F FIRMWARE STATE MACHINE

firmware/src/main.cpp · Feather M0 · Arduino SAMD core



POWER-ON / RESET
cold start: record_seq = 0
watchdog reset: record_seq kept

BOOT · setup()
1. read reset cause (watchdog?) → SOH bit
2. restore retained state if magic = 0x5C002702
3. watchdog on, 16.384 s
4. latch D11 and A3 HIGH (OFF) **before** pinMode OUTPUT; D13 LED off
5. ADC 12-bit; DS18B20 at 12-bit resolution
6. PCF8523 begin → RTC_LOST if unset
7. SD.begin(CS D10)
8. RFM95: reset pulse D4, 915.0 MHz, +11 dBm, regs 0x98 / 0xC4 / 0x04
9. SOH = WATCHDOG_RESET · RTC_UNSET · SD_INIT_FAIL · LORA_INIT_FAIL
10. 8 s USB window: accept `T<unix-epoch>` → set PCF8523 (set_rtc.py); none on battery
11. print the `boot:` line

SAMPLE + LOG · at :00 and :30 UTC
1. D11 LOW, wait 500 ms for the sensor to settle
2. battery A7 → mode NORMAL / CONSERVE / CRITICAL
3. DS18B20 (750 ms) · else TEMP_TIMEOUT
4. turbidity A1 (skipped in CRITICAL)
5. D11 HIGH (off)
6. audio if hour ∈ {0, 8, 16} and NORMAL · **capture not written (G7)**
7. flags · record_seq + 1 · CSV row → /DATA/YYYYMMDD.CSV (retry once → SD_RETRY)
8. hold the reading for the next slot

STANDBY SLEEP
watchdog off · radio sleep · LowPower.deepSleep(Δt)
SAMD21 RTC alarm (1 s resolution) · USB drops
≈ 0.5 mA floor (estimate)

WAKE
re-read PCF8523 · woke early → sleep the rest
clock jumped back → re-plan from the new time
watchdog re-armed · both gates forced OFF
no RTC → fallback clock advanced by hand

TRANSMIT LATEST READING
encode 30 B + CRC-16 · pet the watchdog
copies: NORMAL 3 · CONSERVE 1 (slot 0 only) · CRITICAL 0
gaps 2 s then 4 s · send blocks ≤ 2 s (≈ 0.56 s on air)
never waits for an ACK · missed slot is skipped
nothing sent if LoRa init failed or no reading yet
OTA: listen window after the send (planned)

PLAN NEXT EVENT
next = earliest of (next :00 / :30, next TX slot)
scout_next_event_epoch() · pure, unit-tested

J POWER MODES

battery read first, every sample

Mode	Battery	Senses	Transmits
NORMAL	≥ 3200 mV	temp, turbidity, audio	every slot, 3 copies
CONSERVE	3100–3199 mV	temp, turbidity · no audio	slot 0 only (1 a day), 1 copy · POWER_CONSERVE
CRITICAL	< 3100 mV	temp + logging only	none · BATT_LOW_SKIP_TX

Thresholds are provisional (config.h). The Li-$SOCl_2$ curve is flat, so these mostly trip near end of life. (20)

L SOFTWARE GAPS + VERIFICATION

runbook §1–2 (10-19)

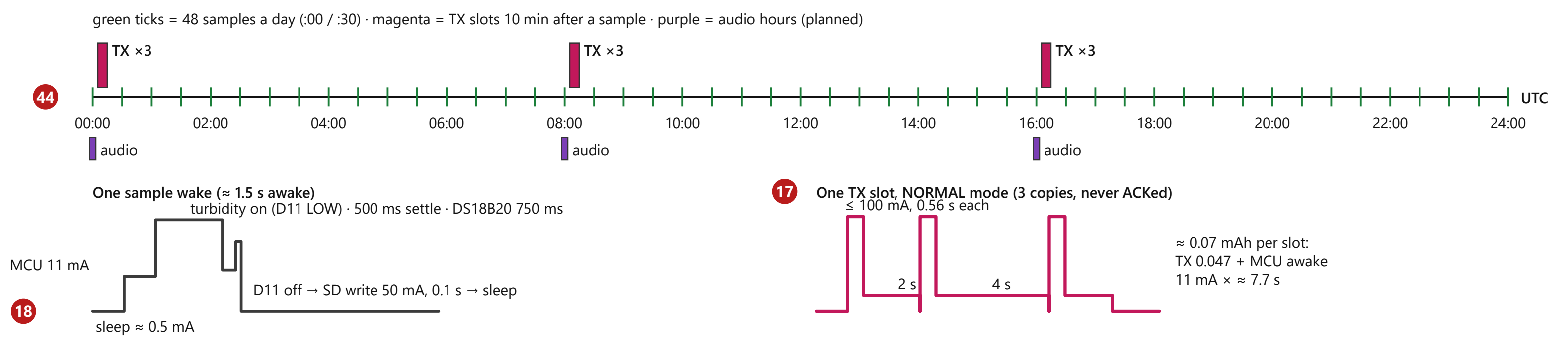
#	Gap	Blocks
G1	**Fixed in code** (PR #186): clock set over USB at boot · confirm RTC_LOST clears (SCO-201)	Stage 1, 4, 5
G2	Design agreed: S3 is the shore station, WiFi push · back end open (SCO-120) · S3 firmware not written (SCO-24)	Stage 4b–4c
G3	Rev A KiCad says DS18B20 on D5; it's D12	none if wired from Rev B
G4	**Fixed in code**: /DATA/YYYYMMDD.CSV, case-insensitive importer · confirm with a real card (SCO-202)	card retrieval
G5	**Fixed in code**: bench receiver feather_m0_rx + decode_frames.py (SCO-155)	Stage 4a
G6	No test build to force a watchdog reset	Stage 5
G7	Audio capture not implemented	Stage 7
OTA	ADR-0008 accepted, protocol not designed (SCO-197)	post-MVP

VERIFIED Stage 0: 20 native firmware tests · target build RAM 19.4% / flash 25.4% · 64 shore tests · 86 analytics tests · packet contract OK (30 B).
UNVERIFIED every driver, the bench tools, the radio registers, sleep current, range. Nothing has met a real sensor or real water yet.

(47) (27) (45)

G 24-HOUR SCHEDULE + CURRENT PROFILE

config.h: SAMPLE_INTERVAL_S 1800 · TX_SLOTS_SOD {600, 29400, 58200} · AUDIO_HOURS_UTC {0, 8, 16}



H LoRa PACKET · 30 BYTES ON THE WIRE

firmware/lib/scout_packet ⇄ shore/scout_shore/packet.py · CI: check_packet_contract.py

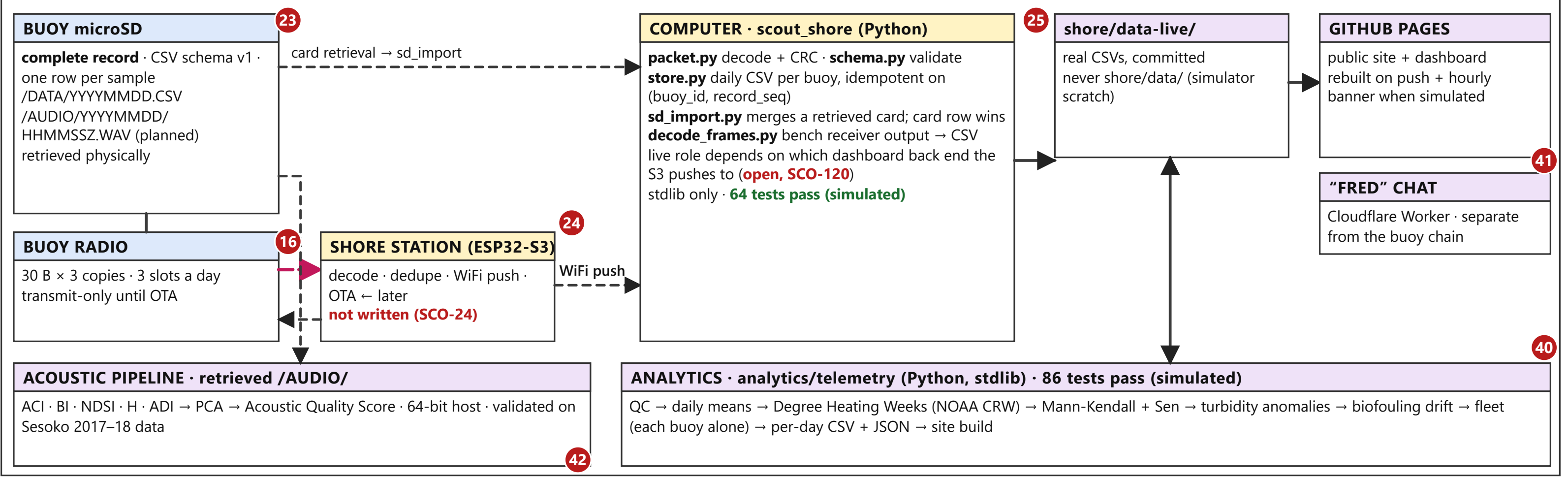
0	1	3	7	11	13	15	17	21	23	24	25	26	27	28	30
ver	buoy_id	timestamp (UTC epoch s)	record_seq	temp_c ×100	turbidity_adc	battery_mv	uptime_s	flags	soh	audio	maj	min	pat	CRC-16	
u8	u16	u32	u32	i16	u16	u16	u32	u16	u8	u8	u8	u8	u8	u16	

(15)

Little-endian. Bytes 0–27 body, 28–29 CRC-16/CCITT-FALSE (poly 0x1021, init 0xFFFF). On air RadioHead adds a 4-byte header (to, from, id, flags). The 82-byte figure in the EDD is the payload *ceiling*, not today's size.
flags bit 0 SD_RETRY · 1 TEMP_TIMEOUT · 2 TURBIDITY_RANGE · 3 BATT_LOW_SKIP_TX · 4 RTC_LOST · 5 POWER_CONSERVE **soh** bit 0 WATCHDOG_RESET · 1 RTC_UNSET · 2 SD_INIT_FAIL · 3 LORA_INIT_FAIL
Not in the packet (SD card only): turbidity_v, turbidity_ntu, audio_file. CSV v1 columns: schema_version, buoy_id, timestamp_utc, record_seq, temp_c, turbidity_adc, turbidity_v, turbidity_ntu, battery_v, uptime_s, audio_file, flags, soh, fw_version.

K DATA PATH · BUOY → SHORE → ANALYTICS → PUBLIC

what runs where, and in what language



BUOY microSD
complete record · CSV schema v1 · one row per sample
/DATA/YYYYMMDD.CSV
/AUDIO/YYYYMMDD/HHMMSSZ.WAV (planned)
retrieved physically

BUOY RADIO
30 B × 3 copies · 3 slots a day
transmit-only until OTA

SHORE STATION (ESP32-S3)
decode · dedupe · WiFi push · OTA ← later
not written (SCO-24)

COMPUTER · scout_shore (Python)
packet.py decode + CRC · **schema.py** validate
store.py daily CSV per buoy, idempotent on (buoy_id, record_seq)
sd_import.py merges a retrieved card; card row wins
decode_frames.py bench receiver output → CSV
live role depends on which dashboard back end the S3 pushes to (**open, SCO-120**)
stdlib only · **64 tests pass (simulated)**

shore/data-live/
real CSVs, committed
never shore/data/ (simulator scratch)

GITHUB PAGES
public site + dashboard rebuilt on push + hourly banner when simulated

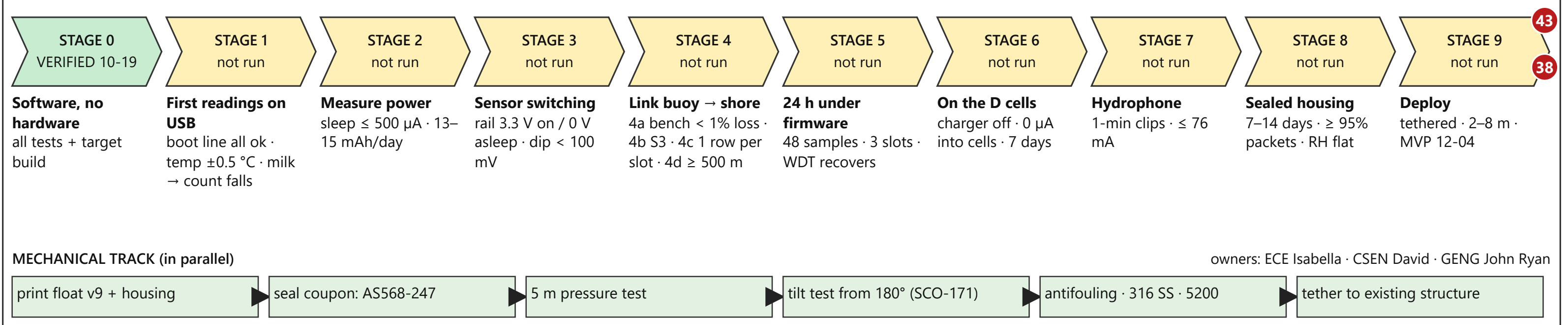
"FRED" CHAT
Cloudflare Worker · separate from the buoy chain

ACOUSTIC PIPELINE · retrieved /AUDIO/
ACI · BI · NDSI · H · ADI → PCA → Acoustic Quality Score · 64-bit host · validated on Sesoko 2017–18 data

ANALYTICS · analytics/telemetry (Python, stdlib) · 86 tests pass (simulated)
QC → daily means → Degree Heating Weeks (NOAA CRW) → Mann-Kendall + Sen → turbidity anomalies → biofouling drift → fleet (each buoy alone) → per-day CSV + JSON → site build

R BUILD, TEST + DEPLOY WORKFLOW

build-and-test plan stages · each stage's PASS is the next stage's precondition



STAGE 0 VERIFIED 10-19	STAGE 1 not run	STAGE 2 not run	STAGE 3 not run	STAGE 4 not run	STAGE 5 not run	STAGE 6 not run	STAGE 7 not run	STAGE 8 not run	STAGE 9 not run
Software, no hardware all tests + target build	**First readings on USB** boot line all ok · temp ±0.5 °C · milk → count falls	**Measure power** sleep ≤ 500 µA · 13–15 mAh/day	**Sensor switching** rail 3.3 V on / 0 V asleep · dip < 100 mV	**Link buoy → shore** 4a bench < 1% loss · 4b S3 · 4c 1 row per slot · 4d ≥ 500 m	**24 h under firmware** 48 samples · 3 slots · WDT recovers	**On the D cells** charger off · 0 µA into cells · 7 days	**Hydrophone** 1-min clips · ≤ 76 mA	**Sealed housing** 7–14 days · ≥ 95% packets · RH flat	**Deploy** tethered · 2–8 m · MVP 12-04

MECHANICAL TRACK (in parallel)

owners: ECE Isabella · CSEN David · GENG John Ryan

print float v9 + housing → seal coupon: AS568-247 → 5 m pressure test → tilt test from 180° (SCO-171) → antifouling · 316 SS · 5200 → tether to existing structure

LEGEND

- battery 3.6 V
- 3.3 V rail / switched rails
- ground
- 1-Wire data
- analog signal
- I^{2}C
- GPIO on/off control
- RF / LoRa
- data / firmware flow
- planned or physical-retrieval path
- cable (mechanical view)
- mooring line

(0) note number (see NOTES)
IF-n interface tag (see register S)
† proposed value, confirm on the bench

🟩 VERIFIED
🟨 UNVERIFIED / OPEN
🟥 GAP · NOT BUILT

▨ electrical part ▨ power
▨ firmware ▨ shore station
▨ data / analytics ▨ mechanical

E FEATHER M0 PIN MAP

Pin	Signal	Type	Note
D12	DS18B20 DQ	1-Wire	4.7 kΩ to 3V3
A1	Turbidity SIG	analog	3.3 V, no divider
D11	Q1 gate (turbidity)	GPIO out	LOW = on
A3	Q2 gate (audio)	GPIO out	LOW = on
A2	Preamp out	analog	Stage 7
A7 / D9	Battery ÷ 2	analog	onboard 100 k / 100 k
D10	microSD CS	SPI CS	Adalogger
D8 · D4 · D3	RFM95 CS · RST · IRQ	SPI + GPIO	on-board, fixed
SCK MOSI MISO	radio + SD	SPI	shared
SDA · SCL	PCF8523 0x68 · SHT40 0x44	I^{2}C	shared
D13	red LED	GPIO	forced off
BAT · GND	pack via JST-PH	power	charger removed
D5 · D6	free	—	D6 kept for radio IO1
A0 A4 A5	spare	—	D0 / D1 = Serial1

(9)

NOTES · numbers match the red balloons on every sheet

(1) Values marked † are proposed, not yet measured. Confirm each on the bench before soldering (Rev B drawing set SCOUT-E-100).

(2) **Remove the Feather's MCP73831 charger before any primary Li-$SOCl_2$ cell is connected.** It charges whatever is on BAT whenever USB is present, and charging a primary cell is a fire hazard. Bench work uses a current-limited 3.6 V supply into BAT.

(3) Cells in **parallel only**: series would be 7.2 V, too high for the 3.3 V regulator and the RFM95. One LM66100 ideal diode per cell stops one cell charging the other; F1 limits a short; P1 is keyed so it can't go in backwards.

(4) The cell must stay ≥ 3.35 V during a 100 mA transmit burst (AP2112K dropout ≈ 250 mV, estimated).

(5) Q1 and Q2 are high-side P-FETs: **gate LOW = sensor ON**. The 100 kΩ pull-ups keep sensors off through boot and reset; firmware latches the gate HIGH before making it an output and re-asserts OFF on every wake (SCO-186).

(6) **Turbidity polarity:** a higher ADC count means **clearer** water. The front end must stay non-inverting; analytics flag downward excursions. Bench check: add milk, the count must fall.

(7) The Grove board runs at 3.3 V and its SIG goes straight to A1, no divider. The buoy logs raw counts; NTU is computed downstream only once a calibration curve exists (SCO-12).

(8) **Open conflict:** Rev B draws the turbidity board inside the electronics housing; the Sensor Pod Packing Budget puts it in the pod's dry chamber. Decide before the gland count is fixed.

(9) DS18B20 data is on **D12** with R1 4.7 kΩ to 3V3. The Rev A KiCad schematic still says D5 (gap G3); wire from this set.

(10) Shared SPI: SD (CS D10) and radio (CS D8), only one LOW at a time. Shared I^{2}C: PCF8523 0x68 and SHT40 0x44, pull-ups already on the boards. If SD and radio work alone but not together, suspect chip select first.

(11) Standby wake comes from the SAMD21's own RTC (32.768 kHz crystal). The PCF8523 is the battery-backed wall clock, re-read after every wake, so drift never accumulates. PCF8523 INT1 is not wired and D5 is free (SCO-196).

(12) **Never power the RFM95 without an antenna.** Deployed antenna ≤ 2.15 dBi and conducted power ≤ 11.6 dBm (FCC ID 2ASEORFM95C, DTS); firmware uses +11 dBm. The housing is engraved "Contains FCC ID: 2ASEORFM95C".

(13) Modem: 915.0 MHz · BW 500 kHz · SF12 · CR 4/8 · explicit header · CRC on. Registers 0x1D = 0x98, 0x1E = 0xC4, 0x26 = 0x04 are computed from the SX1276 map, not bench-verified (SCO-98); fallback SF11 is 0x1E = 0xB4. BW 500 kHz is what 47 CFR §15.247(a)(2) requires.

(14) The shore receiver must match those three bytes exactly and apply the SX1276 500 kHz erratum: reg 0x36 ← 0x02, reg 0x3A ← 0x64 (receive side only).

(15) Packet: 30 bytes (28-byte body + CRC-16/CCITT-FALSE), little-endian, plus RadioHead's 4-byte header on air; ≈ 559 ms airtime. Firmware and shore decoder are checked against each other in CI (scripts/check_packet_contract.py). Change both and bump SCOUT_PACKET_VERSION.

(16) Delivery is blind repetition, never acknowledged: NORMAL 3 copies (gaps 2 s, then 4 s), CONSERVE 1 copy in slot 0 only, CRITICAL none. Shore deduplicates on (buoy_id, record_seq). A missed slot is skipped, not retried; the SD card holds the complete record.

(17) The power budget counts 3 transmissions a day with the MCU awake only while sending; firmware sends 3 slots × 3 copies = 9 and stays awake ≈ 7.7 s per slot. That is ≈ 0.07 mAh per slot, ≈ 0.21 mAh/day against the 0.05 budgeted: +≈ 0.16 mAh/day, about 1% of the budget. Re-run the budget with Stage 2 measurements.

(18) Sleep floor ≈ 501 µA (estimate: 300 µA board + ≈ 200 µA SD idle + 1 µA RTC) is ≈ 90% of the 13.3 mAh/day budget. Measure it first (SCO-23) and record it in facts.md.

(19) Battery life: 2 × 17 Ah × 70% derating = 23.8 Ah usable. Sensors only (13.3 mAh/day) ≈ 4.9 yr; with hydrophone, best estimate (30.3 mAh/day) ≈ 2.1 yr; pessimistic (55.1 mAh/day) ≈ 14 months. Design target 12 months.

(20) Li-$SOCl_2$ discharge is flat, so the 3200 / 3100 mV mode thresholds are provisional and voltage is a poor fuel gauge (SCO-83). On USB, A7 reads the charger output, not a cell.

(21) Watchdog 16.384 s (SAMD21 maximum): off during standby, re-armed every wake, petted between transmit copies. record_seq lives in .noinit RAM (magic 0x5C002702), so it survives a watchdog reset but not a full power loss.

(22) **Set the clock before every deployment.** For ≈ 8 s after a reset, with USB connected, the firmware accepts T<unix-epoch> and writes the PCF8523; firmware/tools/set_rtc.py sends it (gap G1, fixed in code by PR #186, SCO-201). Confirm on hardware that RTC_LOST clears. An unset clock gives wrong timestamps **and fires the TX slots at the wrong times**.

(23) SD layout (gap G4, fixed in code, SCO-202): /DATA/YYYYMMDD.CSV and /AUDIO/YYYYMMDD/HHMMSSZ.WAV, FAT 8.3 names in upper case, no buoy_id in the name (it is in every row). The card importer matches .csv in any case; shore-side files keep <buoy_id>_<YYYYMMDD>.csv. Confirm with a real card.

(24) **Shore design agreed 2026-10-19 (gap G2):** the ESP32-S3 + LoRa FeatherWing *is* the shore station. It runs FreeRTOS, receives, validates and deduplicates, pushes to the dashboard over WiFi, and later sends OTA and settings. **Open:** which dashboard back end receives the push; today's GitHub Pages site has no endpoint, and Catherine's React + SQL dashboard is to be reviewed first (SCO-120). S3 firmware not written (SCO-24, due 2026-10-30).

(25) Audio is never sent over LoRa. Clips stay in /AUDIO/ on the SD card and are retrieved physically (≈ 35 MB/day at 22.05 kHz). Capture code is not written yet (gap G7).

(26) Hydrophone bias: AUD_3V3 → R9 100 Ω / C10 → R4 1.1 kΩ → tip ≈ 2.46 V at 0.7 mA (minimum 2 V). Preamp gain ≈ 231 (47 dB) turns ≈ 2.5 mV of reef sound into ≈ 0.6 V at A2, centred on 1.65 V; low-pass ≈ 10.6 kHz anti-alias. If it clips, change R10 to 10 kΩ.

(27) Over-the-air updates (ADR-0008, accepted 2026-10-16, not designed yet): listen window after each send (RX ≈ 10–12 mA, not in the budget), image chunks staged on SD with per-chunk ACK and resume, authenticated image, rollback if the new image fails to check in, shore TX within the grant. A ≈ 66.6 KB image is ≈ 270 frames ≈ 13 min on air.

(28) SHT40 internal humidity is the leak detector: near-zero RH in a sealed bay, so a rise means a leak. It is on the BOM; firmware and schema columns are not added yet (needs a schema_version bump).

(29) Electronics housing: 4.4 in ID × 5 in (111.8 × 127 mm), flange style, static face seal on an AS568-247 O-ring (ID 117.07 × CS 3.53 mm) with every fastener outside the ring. Confirm the ring against a supplier table (SCO-137). Never seal an unverified board (Stage 8).

(30) All cables enter through glands in the top cap: hydrophone, antenna, DS18B20, turbidity. L3 has 4 × Ø9 mm cable holes. Glands come from lab stock; gland positions in the cap are not drawn yet. The 3.5 mm plug (≈ 6 mm) must pass its gland or be fitted after it.

(31) Packaging option C (2026-10-15): Ø104 mm printed discs on 3 pillars Ø7 at R47.5 (90°/210°/330°) with Ø3 pins or M3 rod. Three layers use 93 of 127 mm, leaving ≈ 34 mm for cable bend. Stack height ≈ 30 mm is unverified, and whether the lid uses part of the 127 mm is not recorded.

(32) The D cells lie on their side in L1 (Ø33.4 × 61.6 mm, ≈ 90–100 g each), vents facing away from the electronics.

(33) Float v9 (2026-10-16): printed PETG, 992 g, no foam, no ballast stem, no lead, with a cable guide channel through the float. At 2.87 kg: draft 2.56 in, freeboard 3.94 in, GM 1.47 in; self-rights from every angle, but GZ is only ≈ 1 mm near 178°. Tilt test from 180° pending (SCO-171).

(34) Stability masses are estimates (battery 250 g, epoxy/coating 330 g, mooring hardware 220 g, pod 217 g, hydrophone 150 g). Replace them with weighed values and re-run run_printed_float.py.

(35) Sensor pod hangs ≈ 4 in below the keel (study value). The dry chamber holds electronics; the flood chamber shades the probe from light. Pod O-ring: Everbilt 2-5/8 in OD × 3/16 in CS. Turbidity leads are epoxy-potted, not O-ring sealed.

(36) Mooring: tether to an existing authorized structure (permitting path B), no anchor of our own. Line: 3/8 in 3-strand nylon, scope TBD (SCO-17). Attachment hardware on the printed float is undecided. Fallback at an unmarked site: one mushroom anchor beside, never on, coral.

(37) Corrosion: 316 stainless hardware with 3M 5200 over exposed bolt heads and joints. Antifouling: Sea Hawk Smart Solution (copper-free; copper is toxic to coral). Fouling drift is screened in analytics (drift.py).

(38) Deployment depth 2–8 m. Pressure-test the sealed housing to 5 m water-equivalent before deployment. Survival design set: d 2 m · H 1.2 m · T 6 s · current 0.8 m/s · wind 22 m/s.

(39) Handling: set the buoy down on its base, never on a sidewall.

(40) Analytics: QC (completeness, gross range, QARTOD flat-line and rate-of-change) → daily means with a coverage floor → NOAA CRW Degree Heating Weeks (MMM + 1 °C, 12 weeks) → Mann-Kendall (Hamed–Rao) + Sen slope → turbidity anomalies (modified z-score) → biofouling drift. Each buoy is analysed alone, with its own MMM.

(41) The dashboard publishes from shore/data-live/, never shore/data/. Simulated data carries a banner. GitHub Pages rebuilds on push to main and hourly.

(42) Acoustic pipeline (retrieved SD cards): ACI, BI, NDSI, H, ADI → PCA → Acoustic Quality Score, on a 64-bit host (Pi OS aarch64 or a laptop).

(43) MVP target 2026-12-04 (hard limit 12-11): the buoy floats and self-rights, takes one temperature and one turbidity reading, and a packet reaches shore.

(44) Samples every 30 min at :00 / :30 UTC. Transmit slots 00:10, 08:10, 16:10 UTC, 10 min after a sample so the radio never overlaps a sensor read. Audio hours 00 / 08 / 16 UTC. Change slot times only in config.h.

(45) Firmware status: Stage 0 verified 2026-10-19 (20 native tests, target build RAM 19.4% / flash 25.4%), recorded before PR #186 added the bench tooling. The drivers and bench tools have never run on hardware; everything downstream is verified on simulated data only.

(46) Sources: repo main @ 455f709 (2026-10-19, through PR #187: runbook rewrite, bench readiness, agreed shore design); Rev B drawing set SCOUT-E-100 (2026-10-12); power budget update 2026-10-05; packaging option C (2026-10-15); printed-float v9 study (2026-10-16); facts.md.

(47) Bench builds (PR #186), never deploy: **feather_m0_bench** samples and transmits every minute and prints a warning at boot; **feather_m0_rx** turns a second Feather M0 into a bench receiver (erratum applied) that prints len, RSSI, SNR and hex per frame for shore/scripts/decode_frames.py. Reflash the normal build before sealing.

PROJECT
S.C.O.U.T. · Santa Clara Oceanic Utilities Transmitter
Nearshore reef monitoring buoy · Santa Clara University Senior Design 2026–27

TITLE
Integrated system: mechanical · electrical · firmware · data

DRAWING NO. **SCOUT-S-000**	SHEET **1 of 4 (master)**
REVISION **A (draft) · first issue**	DATE **2026-10-19**
DRAWN **I. Rodriguez (ECEN)**	CHECKED **— (S. Krishnan to review)**
MECH SOURCE J. R. Myrdal (GENG) CAD	FW / SHORE SOURCE D. Chousal (CSEN) repo
PAPER · SCALE 36 × 24 in (ARCH D) · NTS	SUPERSEDES nothing · companion to SCOUT-E-100 Rev B

S.C.O.U.T. · SYSTEM BLOCK DIAGRAM · SCOUT-S-000

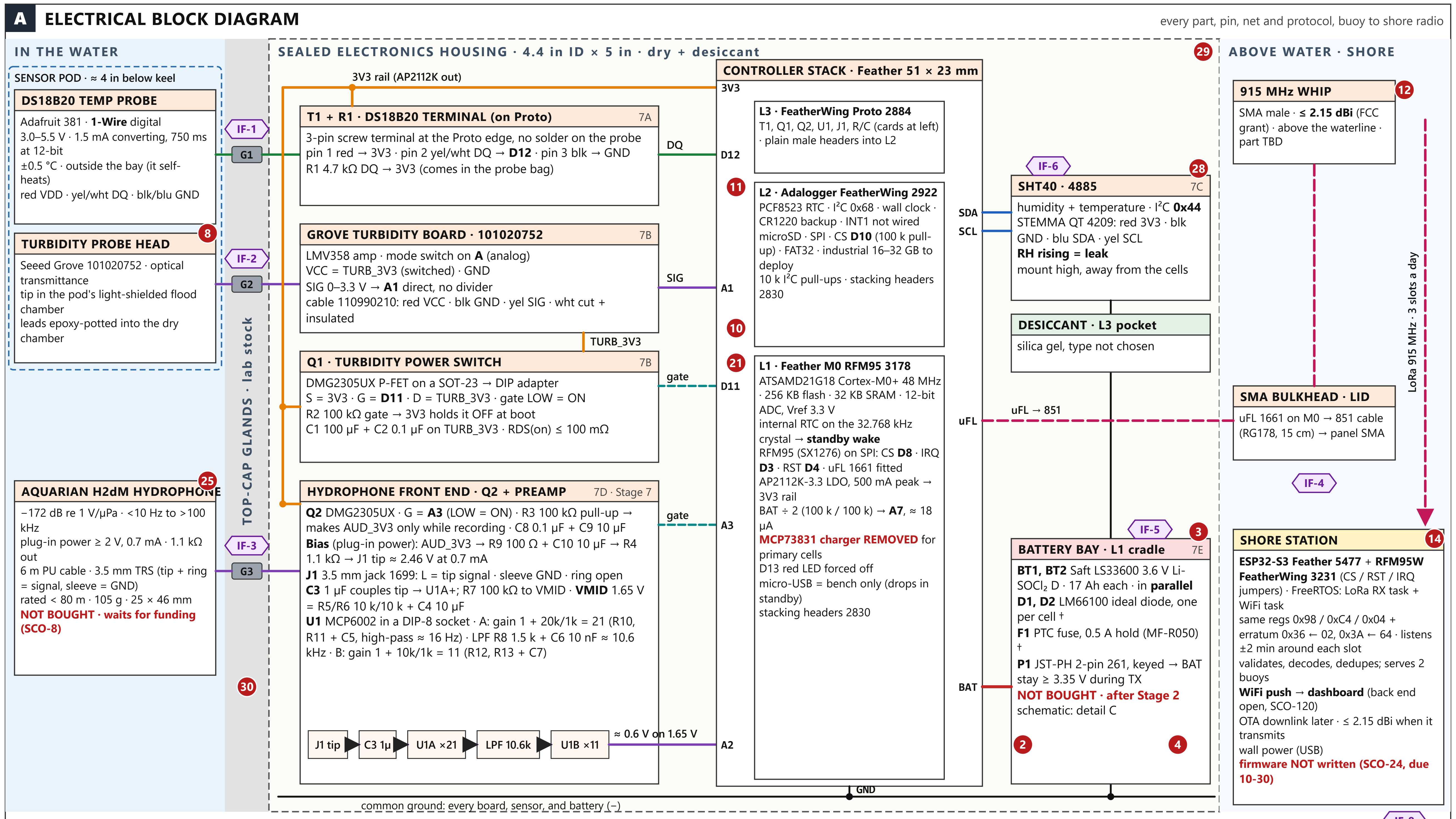
Electrical hardware

zoom of sheet 1 · sections A–E · same note numbers

PRELIMINARY · NOT FOR FINAL BUILD

A ELECTRICAL BLOCK DIAGRAM

every part, pin, net and protocol, buoy to shore radio


IN THE WATER
SENSOR POD · ≈ 4 in below keel
DS18B20 TEMP PROBE
Adafruit 381 · **1-Wire** digital
3.0–5.5 V · 1.5 mA converting, 750 ms at 12-bit
±0.5 °C · outside the bay (it self-heats)
red VDD · yel/wht DQ · blk/blu GND
TURBIDITY PROBE HEAD
8
Seeed Grove 101020752 · optical transmittance
tip in the pod's light-shielded flood chamber
leads epoxy-potted into the dry chamber
AQUARIAN H2dM HYDROPHONE
25
−172 dB re 1 V/μPa · <10 Hz to >100 kHz
plug-in power ≥ 2 V, 0.7 mA · 1.1 kΩ out
6 m PU cable · 3.5 mm TRS (tip + ring = signal, sleeve = GND)
rated < 80 m · 105 g · 25 × 46 mm
NOT BOUGHT · waits for funding (SCO-8)
TOP-CAP GLANDS · lab stock
IF-1
G1
IF-2
G2
IF-3
G3
30
SEALED ELECTRONICS HOUSING · 4.4 in ID × 5 in · dry + desiccant
29
3V3 rail (AP2112K out)
T1 + R1 · DS18B20 TERMINAL (on Proto)
7A
3-pin screw terminal at the Proto edge, no solder on the probe
pin 1 red → 3V3 · pin 2 yel/wht DQ → **D12** · pin 3 blk → GND
R1 4.7 kΩ DQ → 3V3 (comes in the probe bag)
DQ
GROVE TURBIDITY BOARD · 101020752
7B
LMV358 amp · mode switch on **A** (analog)
VCC = TURB_3V3 (switched) · GND
SIG 0–3.3 V → **A1** direct, no divider
cable 110990210: red VCC · blk GND · yel SIG · wht cut + insulated
SIG
TURB_3V3
Q1 · TURBIDITY POWER SWITCH
7B
DMG2305UX P-FET on a SOT-23 → DIP adapter
S = 3V3 · G = **D11** · D = TURB_3V3 · gate LOW = ON
R2 100 kΩ gate → 3V3 holds it OFF at boot
C1 100 μF + C2 0.1 μF on TURB_3V3 · RDS(on) ≤ 100 mΩ
gate
HYDROPHONE FRONT END · Q2 + PREAMP
7D · Stage 7
Q2 DMG2305UX · G = **A3** (LOW = ON) · R3 100 kΩ pull-up → makes AUD_3V3 only while recording · C8 0.1 μF + C9 10 μF
Bias (plug-in power): AUD_3V3 → R9 100 Ω + C10 10 μF → R4 1.1 kΩ → J1 tip ≈ 2.46 V at 0.7 mA
J1 3.5 mm jack 1699: L = tip signal · sleeve GND · ring open
C3 1 μF couples tip → U1A+; R7 100 kΩ to VMID · **VMID** 1.65 V = R5/R6 10 k/10 k + C4 10 μF
U1 MCP6002 in a DIP-8 socket · A: gain 1 + 20k/1k = 21 (R10, R11 + C5, high-pass ≈ 16 Hz) · LPF R8 1.5 k + C6 10 nF ≈ 10.6 kHz · B: gain 1 + 10k/1k = 11 (R12, R13 + C7)
gate
J1 tip
C3 1μ
U1A ×21
LPF 10.6k
U1B ×11
≈ 0.6 V on 1.65 V
CONTROLLER STACK · Feather 51 × 23 mm
3V3
D12
A1
D11
A3
A2
11
10
21
L3 · FeatherWing Proto 2884
T1, Q1, Q2, U1, J1, R/C (cards at left) · plain male headers into L2
L2 · Adalogger FeatherWing 2922
PCF8523 RTC · I²C 0x68 · wall clock · CR1220 backup · INT1 not wired
microSD · SPI · CS **D10** (100 k pull-up) · FAT32 · industrial 16–32 GB to deploy
10 k I²C pull-ups · stacking headers 2830
L1 · Feather M0 RFM95 3178
ATSAMD21G18 Cortex-M0+ 48 MHz · 256 KB flash · 32 KB SRAM · 12-bit ADC, Vref 3.3 V
internal RTC on the 32.768 kHz crystal → **standby wake**
RFM95 (SX1276) on SPI: CS **D8** · IRQ **D3** · RST **D4** · uFL 1661 fitted
AP2112K-3.3 LDO, 500 mA peak → 3V3 rail
BAT ÷ 2 (100 k / 100 k) → **A7**, ≈ 18 μA
MCP73831 charger REMOVED for primary cells
D13 red LED forced off
micro-USB = bench only (drops in standby)
stacking headers 2830
SDA
SCL
uFL
BAT
IF-6
28
SHT40 · 4885
7C
humidity + temperature · I²C **0x44**
STEMMA QT 4209: red 3V3 · blk GND · blu SDA · yel SCL
RH rising = leak
mount high, away from the cells
DESICCANT · L3 pocket
silica gel, type not chosen
uFL → 851
IF-5
3
BATTERY BAY · L1 cradle
7E
BT1, BT2 Saft LS33600 3.6 V Li-SOCl₂ D · 17 Ah each · in **parallel**
D1, D2 LM66100 ideal diode, one per cell †
F1 PTC fuse, 0.5 A hold (MF-R050) †
P1 JST-PH 2-pin 261, keyed → BAT stay ≥ 3.35 V during TX
NOT BOUGHT · after Stage 2
schematic: detail C
2
4
GND
common ground: every board, sensor, and battery (−)
ABOVE WATER · SHORE
915 MHz WHIP
12
SMA male · **≤ 2.15 dBi** (FCC grant) · above the waterline · part TBD
LoRa 915 MHz · 3 slots a day
SMA BULKHEAD · LID
uFL 1661 on M0 → 851 cable (RG178, 15 cm) → panel SMA
IF-4
SHORE STATION
14
ESP32-S3 Feather 5477 + **RFM95W FeatherWing 3231** (CS / RST / IRQ jumpers) · FreeRTOS: LoRa RX task + WiFi task
same regs 0x98 / 0xC4 / 0x04 + erratum 0x36 ← 02, 0x3A ← 64 · listens ±2 min around each slot
validates, decodes, dedupes; serves 2 buoys
WiFi push → **dashboard** (back end open, SCO-120)
OTA downlink later · ≤ 2.15 dBi when it transmits
wall power (USB)
firmware NOT written (SCO-24, due 10-30)
IF-8
1

Inside the stack, no wires to add: SPI SCK / MOSI / MISO is shared by the radio (CS D8) and the SD card (CS D10); I^2C SDA / SCL is shared by the PCF8523 (0x68) and the SHT40 (0x44).
Pins used by the boards: D3, D4, D8 (radio), D9 / A7 (battery), D10 (SD), D13 (LED, off). Free: A0, A4, A5, D5 (old RTC wake), D6 (keep for radio IO1), D0 / D1 (Serial1).

E FEATHER M0 PIN MAP

9

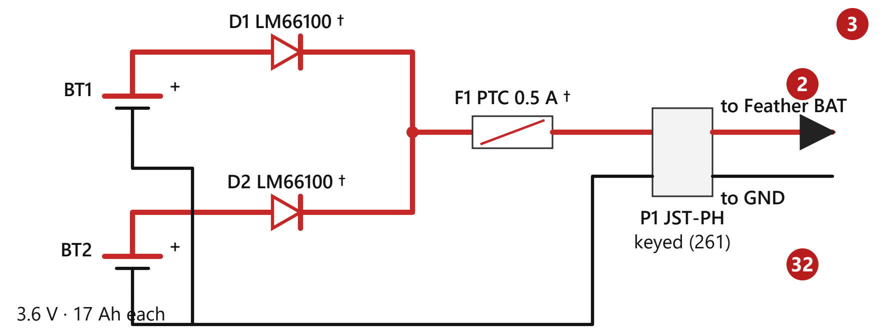
Pin	Signal	Type	Note
D12	DS18B20 DQ	1-Wire	4.7 kΩ to 3V3
A1	Turbidity SIG	analog	3.3 V, no divider
D11	Q1 gate (turbidity)	GPIO out	LOW = on
A3	Q2 gate (audio)	GPIO out	LOW = on
A2	Preamp out	analog	Stage 7
A7 / D9	Battery ÷ 2	analog	onboard 100 k / 100 k
D10	microSD CS	SPI CS	Adalogger
D8 · D4 · D3	RFM95 CS · RST · IRQ	SPI + GPIO	on-board, fixed
SCK MOSI MISO	radio + SD	SPI	shared
SDA · SCL	PCF8523 0x68 · SHT40 0x44	I^2C	shared
D13	red LED	GPIO	forced off
BAT · GND	pack via JST-PH	power	charger removed
D5 · D6	free	—	D6 kept for radio IO1
A0 A4 A5	spare	—	D0 / D1 = Serial1

LEGEND

- ━ battery 3.6 V
- ━ 3.3 V rail / switched rails
- ━ ground
- ━ 1-Wire data
- ━ analog signal
- ━ I^2C
- ╍ GPIO on/off control
- ╍ RF / LoRa
- ━ data / firmware flow
- ╍ planned or physical-retrieval path
- ━ cable (mechanical view)
- ━ mooring line

- ⓪ note number (see NOTES)
- IF-n interface tag (see register S)
- † proposed value, confirm on the bench
- 🟩 VERIFIED
- 🟨 UNVERIFIED / OPEN
- 🟥 GAP · NOT BUILT
- ▨ electrical part
- ▨ firmware
- ▨ data / analytics
- ▨ power
- ▨ shore station
- ▨ mechanical

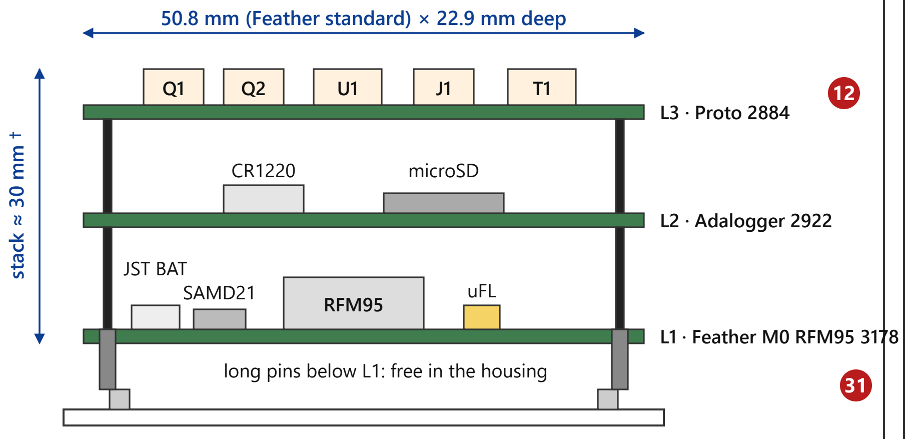
C BATTERY PACK · SCHEMATIC


D1 LM66100 †
3
BT1
+
F1 PTC 0.5 A †
2
to Feather BAT
D2 LM66100 †
to GND
P1 JST-PH
keyed (261)
BT2
+
32
3.6 V · 17 Ah each

Parallel, never series (7.2 V is too high). Each ideal diode stops one cell charging the other; F1 limits a short; P1 cannot go in backwards. Measure P1 before plugging in: +3.5 to 3.7 V. Solder to the tabs only, never the cell body.

B CONTROLLER STACK · SIDE VIEW

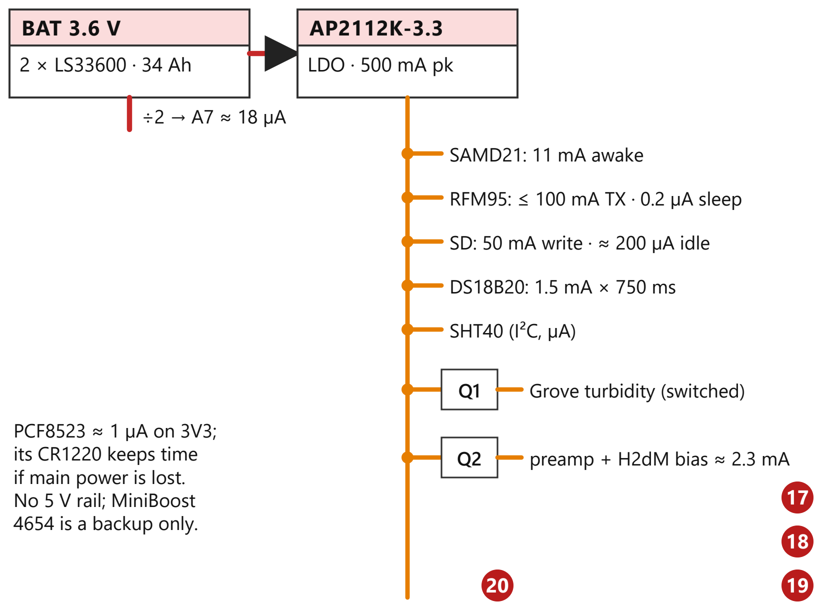
Heights above the housing floor (option C): M0 z 52–60 · Adalogger 61–69 · Proto 70–75 mm. Antenna: bench = spring 4269 on the ANT pad; deployed = uFL 1661 → 851 → SMA bulkhead → whip. Use one, never both.


50.8 mm (Feather standard) × 22.9 mm deep
stack ≈ 30 mm †
Q1
Q2
U1
J1
T1
L3 · Proto 2884
12
CR1220
microSD
L2 · Adalogger 2922
JST BAT
SAMD21
RFM95
uFL
L1 · Feather M0 RFM95 3178
long pins below L1: free in the housing
31

L2 stack tray (3 mm) on 4 × Ø6 × 4 standoffs at (±22.9, ±8.9) mm

D POWER TREE + DAILY ENERGY BUDGET

power budget update 2026-10-05


BAT 3.6 V
2 × LS33600 · 34 Ah
AP2112K-3.3
LDO · 500 mA pk
÷2 → A7 ≈ 18 μA
SAMD21: 11 mA awake
RFM95: ≤ 100 mA TX · 0.2 μA sleep
SD: 50 mA write · ≈ 200 μA idle
DS18B20: 1.5 mA × 750 ms
SHT40 (I²C, μA)
Q1
Grove turbidity (switched)
Q2
preamp + H2dM bias ≈ 2.3 mA
17
18
19
20

PCF8523 ≈ 1 μA on 3V3; its CR1220 keeps time if main power is lost. No 5 V rail; MiniBoost 4654 is a backup only.

Load	mA	s	×/d	mAh/d
SAMD21 awake	11	1.5	48	0.220
DS18B20	1.5	0.75	48	0.015
Turbidity*	71.3	1.0	48	0.951
SD write	50	0.1	48	0.067
RFM95 TX	≤100	0.559	3	0.047
MCU during TX	11	0.559	3	0.005
Sleep floor	0.501	24 h	1	12.02
Total, no audio	**avg ≈ 0.56 mA**			**13.33**
Hydrophone	≈75.7	13.4 min/d		16.9
Total, with audio	**best estimate**			**30.3**

*Budget row is the SEN0189 through a 5 V boost (worst case); the Grove at 3.3 V draws less. Audio row assumed a PCM1808 ADC; the MCP6002 path is lower. Usable 23.8 Ah → ≈ 4.9 yr without audio, ≈ 2.1 yr with.

NOTES · same numbers as sheet 1

① Values marked † are proposed, not yet measured. Confirm each on the bench before soldering (Rev B drawing set SCOUT-E-100).

② **Remove the Feather's MCP73831 charger before any primary Li-$SOCl_2$ cell is connected.** It charges whatever is on BAT whenever USB is present, and charging a primary cell is a fire hazard. Bench work uses a current-limited 3.6 V supply into BAT.

③ Cells in **parallel only**: series would be 7.2 V, too high for the 3.3 V regulator and the RFM95. One LM66100 ideal diode per cell stops one cell charging the other; F1 limits a short; P1 is keyed so it can't go in backwards.

④ The cell must stay ≥ 3.35 V during a 100 mA transmit burst (AP2112K dropout ≈ 250 mV, estimated).

⑧ **Open conflict:** Rev B draws the turbidity board inside the electronics housing; the Sensor Pod Packing Budget puts it in the pod's dry chamber. Decide before the gland count is fixed.

⑨ DS18B20 data is on **D12** with R1 4.7 kΩ to 3V3. The Rev A KiCad schematic still says D5 (gap G3); wire from this set.

⑩ Shared SPI: SD (CS D10) and radio (CS D8), only one LOW at a time. Shared I^2C: PCF8523 0x68 and SHT40 0x44, pull-ups already on the boards. If SD and radio work alone but not together, suspect chip select first.

⑪ Standby wake comes from the SAMD21's own RTC (32.768 kHz crystal). The PCF8523 is the battery-backed wall clock, re-read after every wake, so drift never accumulates. PCF8523 INT1 is not wired and D5 is free (SCO-196).

⑫ **Never power the RFM95 without an antenna.** Deployed antenna ≤ 2.15 dBi and conducted power ≤ 11.6 dBm (FCC ID 2ASEORFM95C, DTS); firmware uses +11 dBm. The housing is engraved “Contains FCC ID: 2ASEORFM95C”.

⑭ The shore receiver must match those three bytes exactly and apply the SX1276 500 kHz erratum: reg 0x36 ← 0x02, reg 0x3A ← 0x64 (receive side only).

⑰ The power budget counts 3 transmissions a day with the MCU awake only while sending; firmware sends 3 slots × 3 copies = 9 and stays awake ≈ 7.7 s per slot. That is ≈ 0.07 mAh per slot, ≈ 0.21 mAh/day against the 0.05 budgeted: +≈ 0.16 mAh/day, about 1% of the budget. Re-run the budget with Stage 2 measurements.

⑱ Sleep floor ≈ 501 μA (estimate: 300 μA board + ≈ 200 μA SD idle + 1 μA RTC) is ≈ 90% of the 13.3 mAh/day budget. Measure it first (SCO-23) and record it in facts.md.

⑲ Battery life: 2 × 17 Ah × 70% derating = 23.8 Ah usable. Sensors only (13.3 mAh/day) ≈ 4.9 yr; with hydrophone, best estimate (30.3 mAh/day) ≈ 2.1 yr; pessimistic (55.1 mAh/day) ≈ 14 months. Design target 12 months.

⑳ Li-$SOCl_2$ discharge is flat, so the 3200 / 3100 mV mode thresholds are provisional and voltage is a poor fuel gauge (SCO-83). On USB, A7 reads the charger output, not a cell.

㉑ Watchdog 16.384 s (SAMD21 maximum): off during standby, re-armed every wake, petted between transmit copies. record_seq lives in .noinit RAM (magic 0x5C002702), so it survives a watchdog reset but not a full power loss.

㉕ Audio is never sent over LoRa. Clips stay in /AUDIO/ on the SD card and are retrieved physically (≈ 35 MB/day at 22.05 kHz). Capture code is not written yet (gap G7).

㉘ SHT40 internal humidity is the leak detector: near-zero RH in a sealed bay, so a rise means a leak. It is on the BOM; firmware and schema columns are not added yet (needs a schema_version bump).

㉙ Electronics housing: 4.4 in ID × 5 in (111.8 × 127 mm), flange style, static face seal on an AS568-247 O-ring (ID 117.07 × CS 3.53 mm) with every fastener outside the ring. Confirm the ring against a supplier table (SCO-137). Never seal an unverified board (Stage 8).

㉚ All cables enter through glands in the top cap: hydrophone, antenna, DS18B20, turbidity. L3 has 4 × Ø9 mm cable holes. Glands come from lab stock; gland positions in the cap are not drawn yet. The 3.5 mm plug (≈ 6 mm) must pass its gland or be fitted after it.

㉛ Packaging option C (2026-10-15): Ø104 mm printed discs on 3 pillars Ø7 at R47.5 (90°/210°/330°) with Ø3 pins or M3 rod. Three layers use 93 of 127 mm, leaving ≈ 34 mm for cable bend. Stack height ≈ 30 mm is unverified, and whether the lid uses part of the 127 mm is not recorded.

㉜ The D cells lie on their side in L1 (Ø33.4 × 61.6 mm, ≈ 90–100 g each), vents facing away from the electronics.

㊻ Sources: repo main @ 455f709 (2026-10-19, through PR #187: runbook rewrite, bench readiness, agreed shore design); Rev B drawing set SCOUT-E-100 (2026-10-12); power budget update 2026-10-05; packaging option C (2026-10-15); printed-float v9 study (2026-10-16); facts.md.

PROJECT
S.C.O.U.T. · Santa Clara Oceanic Utilities Transmitter
Nearshore reef monitoring buoy · Santa Clara University Senior Design 2026–27

TITLE
Electrical hardware

DRAWING NO. **SCOUT-S-000**	SHEET **2 of 4**
REVISION **A (draft) · first issue**	DATE **2026-10-19**
DRAWN **I. Rodriguez (ECEN)**	CHECKED **— (S. Krishnan to review)**
MECH SOURCE J. R. Myrdal (GENG) CAD	FW / SHORE SOURCE D. Chousal (CSEN) repo
PAPER · SCALE 22 × 17 in (ANSI C) · NTS	SUPERSEDES nothing · companion to SCOUT-E-100 Rev B

S.C.O.U.T. · SYSTEM BLOCK DIAGRAM · SCOUT-S-000

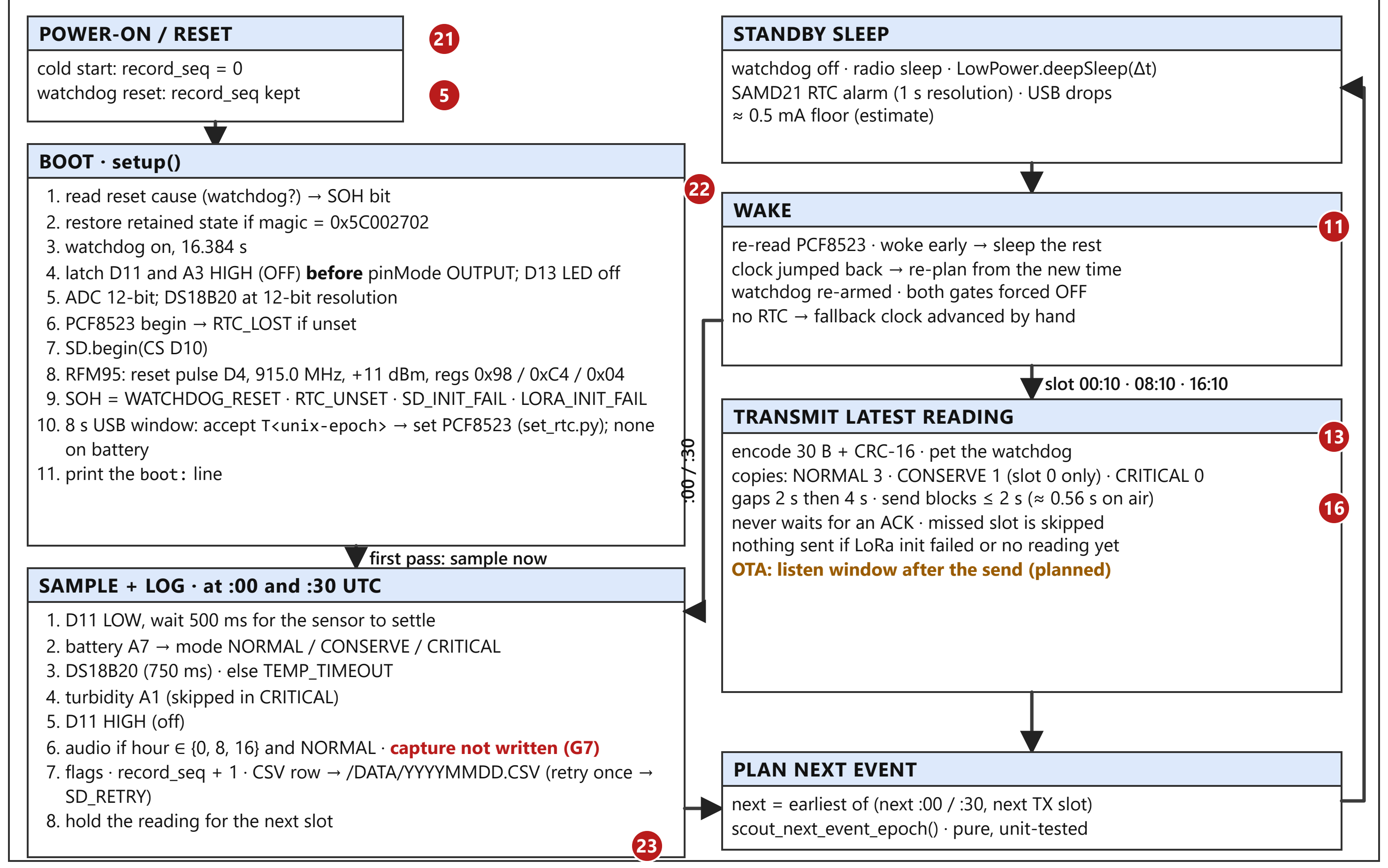
Firmware, radio link and data

zoom of sheet 1 · sections F–L, R · same note numbers

PRELIMINARY · NOT FOR FINAL BUILD

F FIRMWARE STATE MACHINE

firmware/src/main.cpp · Feather M0 · Arduino SAMD core



POWER-ON / RESET ㉑ ⑤

cold start: record_seq = 0
watchdog reset: record_seq kept

BOOT · setup() ㉒

1. read reset cause (watchdog?) → SOH bit
2. restore retained state if magic = 0x5C002702
3. watchdog on, 16.384 s
4. latch D11 and A3 HIGH (OFF) **before** pinMode OUTPUT; D13 LED off
5. ADC 12-bit; DS18B20 at 12-bit resolution
6. PCF8523 begin → RTC_LOST if unset
7. SD.begin(CS D10)
8. RFM95: reset pulse D4, 915.0 MHz, +11 dBm, regs 0x98 / 0xC4 / 0x04
9. SOH = WATCHDOG_RESET · RTC_UNSET · SD_INIT_FAIL · LORA_INIT_FAIL
10. 8 s USB window: accept `T<unix-epoch>` → set PCF8523 (set_rtc.py); none on battery
11. print the `boot:` line

first pass: sample now

SAMPLE + LOG · at :00 and :30 UTC ㉓

1. D11 LOW, wait 500 ms for the sensor to settle
2. battery A7 → mode NORMAL / CONSERVE / CRITICAL
3. DS18B20 (750 ms) · else TEMP_TIMEOUT
4. turbidity A1 (skipped in CRITICAL)
5. D11 HIGH (off)
6. audio if hour ∈ {0, 8, 16} and NORMAL · **capture not written (G7)**
7. flags · record_seq + 1 · CSV row → /DATA/YYYYMMDD.CSV (retry once → SD_RETRY)
8. hold the reading for the next slot

:00 / :30

STANDBY SLEEP

watchdog off · radio sleep · LowPower.deepSleep(Δt)
SAMD21 RTC alarm (1 s resolution) · USB drops
≈ 0.5 mA floor (estimate)

WAKE ⑪

re-read PCF8523 · woke early → sleep the rest
clock jumped back → re-plan from the new time
watchdog re-armed · both gates forced OFF
no RTC → fallback clock advanced by hand

slot 00:10 · 08:10 · 16:10

TRANSMIT LATEST READING ⑬ ⑯

encode 30 B + CRC-16 · pet the watchdog
copies: NORMAL 3 · CONSERVE 1 (slot 0 only) · CRITICAL 0
gaps 2 s then 4 s · send blocks ≤ 2 s (≈ 0.56 s on air)
never waits for an ACK · missed slot is skipped
nothing sent if LoRa init failed or no reading yet
OTA: listen window after the send (planned)

PLAN NEXT EVENT

next = earliest of (next :00 / :30, next TX slot)
scout_next_event_epoch() · pure, unit-tested

J POWER MODES

battery read first, every sample

Mode	Battery	Senses	Transmits
NORMAL	≥ 3200 mV	temp, turbidity, audio	every slot, 3 copies
CONSERVE	3100–3199 mV	temp, turbidity · no audio	slot 0 only (1 a day), 1 copy · POWER_CONSERVE
CRITICAL	< 3100 mV	temp + logging only	none · BATT_LOW_SKIP_TX

Thresholds are provisional (config.h). The Li-$SOCl_2$ curve is flat, so these mostly trip near end of life. ⑳

L SOFTWARE GAPS + VERIFICATION

runbook §1–2 (10-19)

#	Gap	Blocks
G1	**Fixed in code** (PR #186): clock set over USB at boot · confirm RTC_LOST clears (SCO-201)	Stage 1, 4, 5
G2	Design agreed: S3 is the shore station, WiFi push · back end open (SCO-120) · S3 firmware not written (SCO-24)	Stage 4b–4c
G3	Rev A KiCad says DS18B20 on D5; it's D12	none if wired from Rev B
G4	**Fixed in code**: /DATA/YYYYMMDD.CSV, case-insensitive importer · confirm with a real card (SCO-202)	card retrieval
G5	**Fixed in code**: bench receiver feather_m0_rx + decode_frames.py (SCO-155)	Stage 4a
G6	No test build to force a watchdog reset	Stage 5
G7	Audio capture not implemented	Stage 7
OTA	ADR-0008 accepted, protocol not designed (SCO-197)	post-MVP

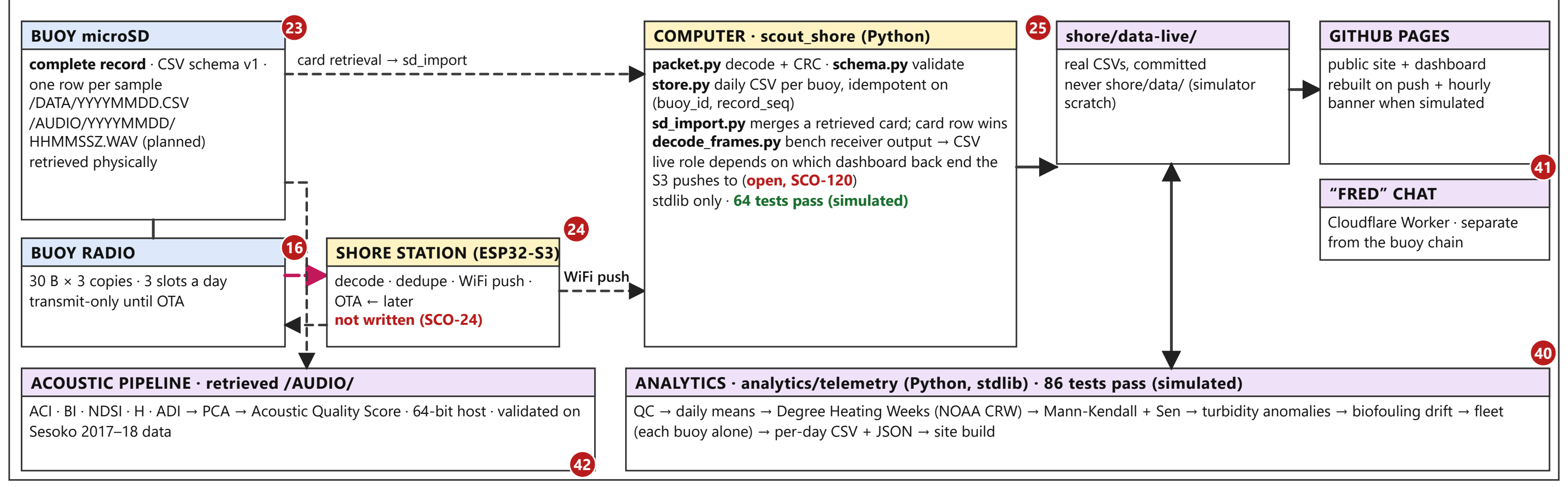
VERIFIED Stage 0: 20 native firmware tests · target build RAM 19.4% / flash 25.4% · 64 shore tests · 86 analytics tests · packet contract OK (30 B).

UNVERIFIED every driver, the bench tools, the radio registers, sleep current, range. Nothing has met a real sensor or real water yet.

㊼ ㉗ ㊺

K DATA PATH · BUOY → SHORE → ANALYTICS → PUBLIC

what runs where, and in what language



BUOY microSD ㉓

complete record · CSV schema v1 · one row per sample
/DATA/YYYYMMDD.CSV
/AUDIO/YYYYMMDD/HHMMSSZ.WAV (planned)
retrieved physically

card retrieval → sd_import

BUOY RADIO ⑯

30 B × 3 copies · 3 slots a day
transmit-only until OTA

SHORE STATION (ESP32-S3) ㉔

decode · dedupe · WiFi push ·
OTA ← later
not written (SCO-24)

WiFi push

COMPUTER · scout_shore (Python) ㉕

packet.py decode + CRC · **schema.py** validate
store.py daily CSV per buoy, idempotent on (buoy_id, record_seq)
sd_import.py merges a retrieved card; card row wins
decode_frames.py bench receiver output → CSV
live role depends on which dashboard back end the S3 pushes to (**open, SCO-120**)
stdlib only · **64 tests pass (simulated)**

shore/data-live/

real CSVs, committed
never shore/data/ (simulator scratch)

GITHUB PAGES ㊶

public site + dashboard
rebuilt on push + hourly
banner when simulated

“FRED” CHAT

Cloudflare Worker · separate from the buoy chain

ACOUSTIC PIPELINE · retrieved /AUDIO/ ㊷

ACI · BI · NDSI · H · ADI → PCA → Acoustic Quality Score · 64-bit host · validated on Sesoko 2017–18 data

ANALYTICS · analytics/telemetry (Python, stdlib) · 86 tests pass (simulated) ㊵

QC → daily means → Degree Heating Weeks (NOAA CRW) → Mann-Kendall + Sen → turbidity anomalies → biofouling drift → fleet (each buoy alone) → per-day CSV + JSON → site build

LEGEND

- battery 3.6 V
- 3.3 V rail / switched rails
- ground
- 1-Wire data
- analog signal
- I^2C
- GPIO on/off control
- RF / LoRa
- data / firmware flow
- planned or physical-retrieval path
- cable (mechanical view)
- mooring line

⓪ note number (see NOTES)

IF-n interface tag (see register S)

† proposed value, confirm on the bench

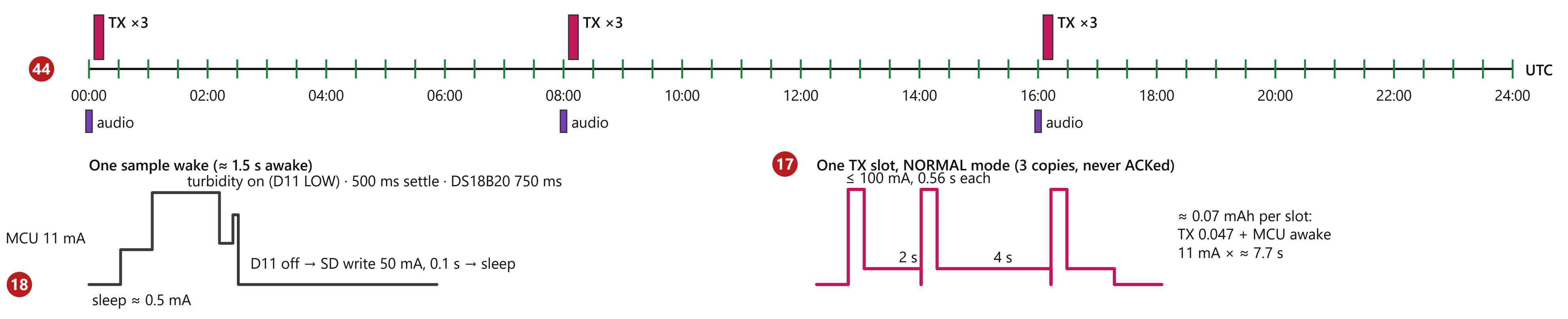
- 🟩 VERIFIED
- 🟨 UNVERIFIED / OPEN
- 🟥 GAP · NOT BUILT

☐ electrical part	☐ power
☐ firmware	☐ shore station
☐ data / analytics	☐ mechanical

G 24-HOUR SCHEDULE + CURRENT PROFILE

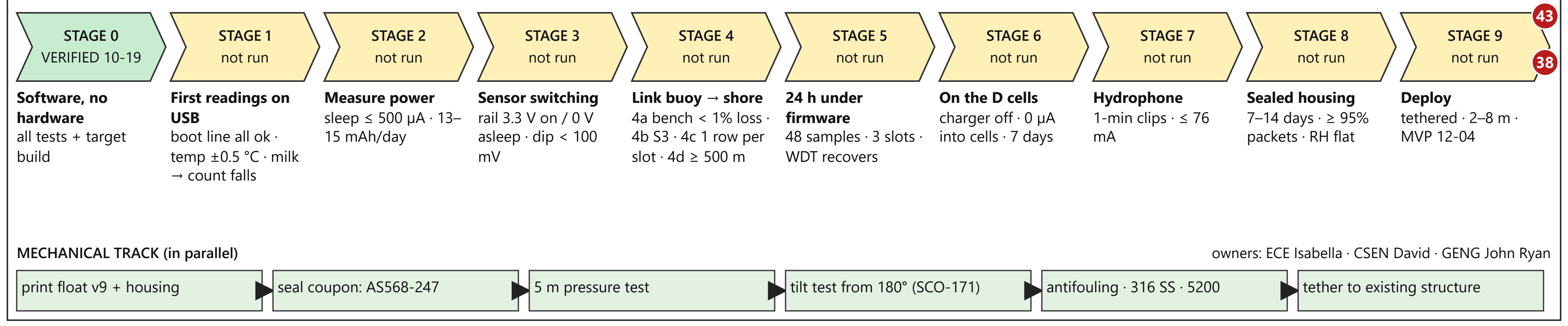
config.h: SAMPLE_INTERVAL_S 1800 · TX_SLOTS_SOD {600, 29400, 58200} · AUDIO_HOURS_UTC {0, 8, 16}

green ticks = 48 samples a day (:00 / :30) · magenta = TX slots 10 min after a sample · purple = audio hours (planned)



R BUILD, TEST + DEPLOY WORKFLOW

build-and-test plan stages · each stage's PASS is the next stage's precondition



STAGE 0 VERIFIED 10-19	STAGE 1 not run	STAGE 2 not run	STAGE 3 not run	STAGE 4 not run	STAGE 5 not run	STAGE 6 not run	STAGE 7 not run	STAGE 8 not run	STAGE 9 not run ㊸ ㊳
Software, no hardware all tests + target build	**First readings on USB** boot line all ok · temp ±0.5 °C · milk → count falls	**Measure power** sleep ≤ 500 µA · 13–15 mAh/day	**Sensor switching** rail 3.3 V on / 0 V asleep · dip < 100 mV	**Link buoy → shore** 4a bench < 1% loss · 4b S3 · 4c 1 row per slot · 4d ≥ 500 m	**24 h under firmware** 48 samples · 3 slots · WDT recovers	**On the D cells** charger off · 0 µA into cells · 7 days	**Hydrophone** 1-min clips · ≤ 76 mA	**Sealed housing** 7–14 days · ≥ 95% packets · RH flat	**Deploy** tethered · 2–8 m · MVP 12-04

MECHANICAL TRACK (in parallel)

owners: ECE Isabella · CSEN David · GENG John Ryan

print float v9 + housing → seal coupon: AS568-247 → 5 m pressure test → tilt test from 180° (SCO-171) → antifouling · 316 SS · 5200 → tether to existing structure

H LoRa PACKET · 30 BYTES ON THE WIRE

firmware/lib/scout_packet ⇄ shore/scout_shore/packet.py · CI: check_packet_contract.py ⑮

Offset	0	1	3	7	11	13	15	17	21	23	24	25	26	27	28
Field	ver	buoy_id	timestamp (UTC epoch s)	record_seq	temp_c ×100	turbidity_adc	battery_mv	uptime_s	flags	soh	audio	maj	min	pat	CRC-16
Type	u8	u16	u32	u32	i16	u16	u16	u32	u16	u8	u8	u8	u8	u8	u16

30

Little-endian. Bytes 0–27 body, 28–29 CRC-16/CCITT-FALSE (poly 0x1021, init 0xFFFF). On air RadioHead adds a 4-byte header (to, from, id, flags). The 82-byte figure in the EDD is the payload *ceiling*, not today's size.

flags bit 0 SD_RETRY · 1 TEMP_TIMEOUT · 2 TURBIDITY_RANGE · 3 BATT_LOW_SKIP_TX · 4 RTC_LOST · 5 POWER_CONSERVE **soh** bit 0 WATCHDOG_RESET · 1 RTC_UNSET · 2 SD_INIT_FAIL · 3 LORA_INIT_FAIL

Not in the packet (SD card only): turbidity_v, turbidity_ntu, audio_file. CSV v1 columns: schema_version, buoy_id, timestamp_utc, record_seq, temp_c, turbidity_adc, turbidity_v, turbidity_ntu, battery_v, uptime_s, audio_file, flags, soh, fw_version.

NOTES · same numbers as sheet 1

⑤ Q1 and Q2 are high-side P-FETs: **gate LOW = sensor ON**. The 100 kΩ pull-ups keep sensors off through boot and reset; firmware latches the gate HIGH before making it an output and re-asserts OFF on every wake (SCO-186).

⑪ Standby wake comes from the SAMD21's own RTC (32.768 kHz crystal). The PCF8523 is the battery-backed wall clock, re-read after every wake, so drift never accumulates. PCF8523 INT1 is not wired and D5 is free (SCO-196).

⑬ Modem: 915.0 MHz · BW 500 kHz · SF12 · CR 4/8 · explicit header · CRC on. Registers 0x1D = 0x98, 0x1E = 0xC4, 0x26 = 0x04 are computed from the SX1276 map, not bench-verified (SCO-98); fallback SF11 is 0x1E = 0xB4. BW 500 kHz is what 47 CFR §15.247(a)(2) requires.

⑮ Packet: 30 bytes (28-byte body + CRC-16/CCITT-FALSE), little-endian, plus RadioHead's 4-byte header on air; ≈ 559 ms airtime. Firmware and shore decoder are checked against each other in CI (scripts/check_packet_contract.py). Change both and bump SCOUT_PACKET_VERSION.

⑯ Delivery is blind repetition, never acknowledged: NORMAL 3 copies (gaps 2 s, then 4 s), CONSERVE 1 copy in slot 0 only, CRITICAL none. Shore deduplicates on (buoy_id, record_seq). A missed slot is skipped, not retried; the SD card holds the complete record.

⑰ The power budget counts 3 transmissions a day with the MCU awake only while sending; firmware sends 3 slots × 3 copies = 9 and stays awake ≈ 7.7 s per slot. That is ≈ 0.07 mAh per slot, ≈ 0.21 mAh/day against the 0.05 budgeted: +≈ 0.16 mAh/day, about 1% of the budget. Re-run the budget with Stage 2 measurements.

⑱ Sleep floor ≈ 501 µA (estimate: 300 µA board + ≈ 200 µA SD idle + 1 µA RTC) is ≈ 90% of the 13.3 mAh/day budget. Measure it first (SCO-23) and record it in facts.md.

⑳ Li-$SOCl_2$ discharge is flat, so the 3200 / 3100 mV mode thresholds are provisional and voltage is a poor fuel gauge (SCO-83). On USB, A7 reads the charger output, not a cell.

㉑ Watchdog 16.384 s (SAMD21 maximum): off during standby, re-armed every wake, petted between transmit copies. record_seq lives in .noinit RAM (magic 0x5C002702), so it survives a watchdog reset but not a full power loss.

㉒ **Set the clock before every deployment.** For ≈ 8 s after a reset, with USB connected, the firmware accepts T<unix-epoch> and writes the PCF8523; firmware/tools/set_rtc.py sends it (gap G1, fixed in code by PR #186, SCO-201). Confirm on hardware that RTC_LOST clears. An unset clock gives wrong timestamps **and fires the TX slots at the wrong times**.

㉓ SD layout (gap G4, fixed in code, SCO-202): /DATA/YYYYMMDD.CSV and /AUDIO/YYYYMMDD/HHMMSSZ.WAV, FAT 8.3 names in upper case, no buoy_id in the name (it is in every row). The card importer matches .csv in any case; shore-side files keep <buoy_id>_<YYYYMMDD>.csv. Confirm with a real card.

㉔ **Shore design agreed 2026-10-19 (gap G2):** the ESP32-S3 + LoRa FeatherWing *is* the shore station. It runs FreeRTOS, receives, validates and deduplicates, pushes to the dashboard over WiFi, and later sends OTA and settings. **Open:** which dashboard back end receives the push; today's GitHub Pages site has no endpoint, and Catherine's React + SQL dashboard is to be reviewed first (SCO-120). S3 firmware not written (SCO-24, due 2026-10-30).

㉕ Audio is never sent over LoRa. Clips stay in /AUDIO/ on the SD card and are retrieved physically (≈ 35 MB/day at 22.05 kHz). Capture code is not written yet (gap G7).

㉗ Over-the-air updates (ADR-0008, accepted 2026-10-16, not designed yet): listen window after each send (RX ≈ 10–12 mA, not in the budget), image chunks staged on SD with per-chunk ACK and resume, authenticated image, rollback if the new image fails to check in, shore TX within the grant. A ≈ 66.6 KB image is ≈ 270 frames ≈ 13 min on air.

㊳ Deployment depth 2–8 m. Pressure-test the sealed housing to 5 m water-equivalent before deployment. Survival design set: d 2 m · H 1.2 m · T 6 s · current 0.8 m/s · wind 22 m/s.

㊵ Analytics: QC (completeness, gross range, QARTOD flat-line and rate-of-change) → daily means with a coverage floor → NOAA CRW Degree Heating Weeks (MMM + 1 °C, 12 weeks) → Mann-Kendall (Hamed–Rao) + Sen slope → turbidity anomalies (modified z-score) → biofouling drift. Each buoy is analysed alone, with its own MMM.

㊶ The dashboard publishes from shore/data-live/, never shore/data/. Simulated data carries a banner. GitHub Pages rebuilds on push to main and hourly.

㊷ Acoustic pipeline (retrieved SD cards): ACI, BI, NDSI, H, ADI → PCA → Acoustic Quality Score, on a 64-bit host (Pi OS aarch64 or a laptop).

㊸ MVP target 2026-12-04 (hard limit 12-11): the buoy floats and self-rights, takes one temperature and one turbidity reading, and a packet reaches shore.

㊹ Samples every 30 min at :00 / :30 UTC. Transmit slots 00:10, 08:10, 16:10 UTC, 10 min after a sample so the radio never overlaps a sensor read. Audio hours 00 / 08 / 16 UTC. Change slot times only in config.h.

㊺ Firmware status: Stage 0 verified 2026-10-19 (20 native tests, target build RAM 19.4% / flash 25.4%), recorded before PR #186 added the bench tooling. The drivers and bench tools have never run on hardware; everything downstream is verified on simulated data only.

㊼ Bench builds (PR #186), never deploy: **feather_m0_bench** samples and transmits every minute and prints a warning at boot; **feather_m0_rx** turns a second Feather M0 into a bench receiver (erratum applied) that prints len, RSSI, SNR and hex per frame for shore/scripts/decode_frames.py. Reflash the normal build before sealing.

㊻ Sources: repo main @ 455f709 (2026-10-19, through PR #187: runbook rewrite, bench readiness, agreed shore design); Rev B drawing set SCOUT-E-100 (2026-10-12); power budget update 2026-10-05; packaging option C (2026-10-15); printed-float v9 study (2026-10-16); facts.md.

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S.C.O.U.T. · Santa Clara Oceanic Utilities Transmitter
Nearshore reef monitoring buoy · Santa Clara University Senior Design 2026–27

TITLE
Firmware, radio link and data

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S.C.O.U.T. · SYSTEM BLOCK DIAGRAM · SCOUT-S-000

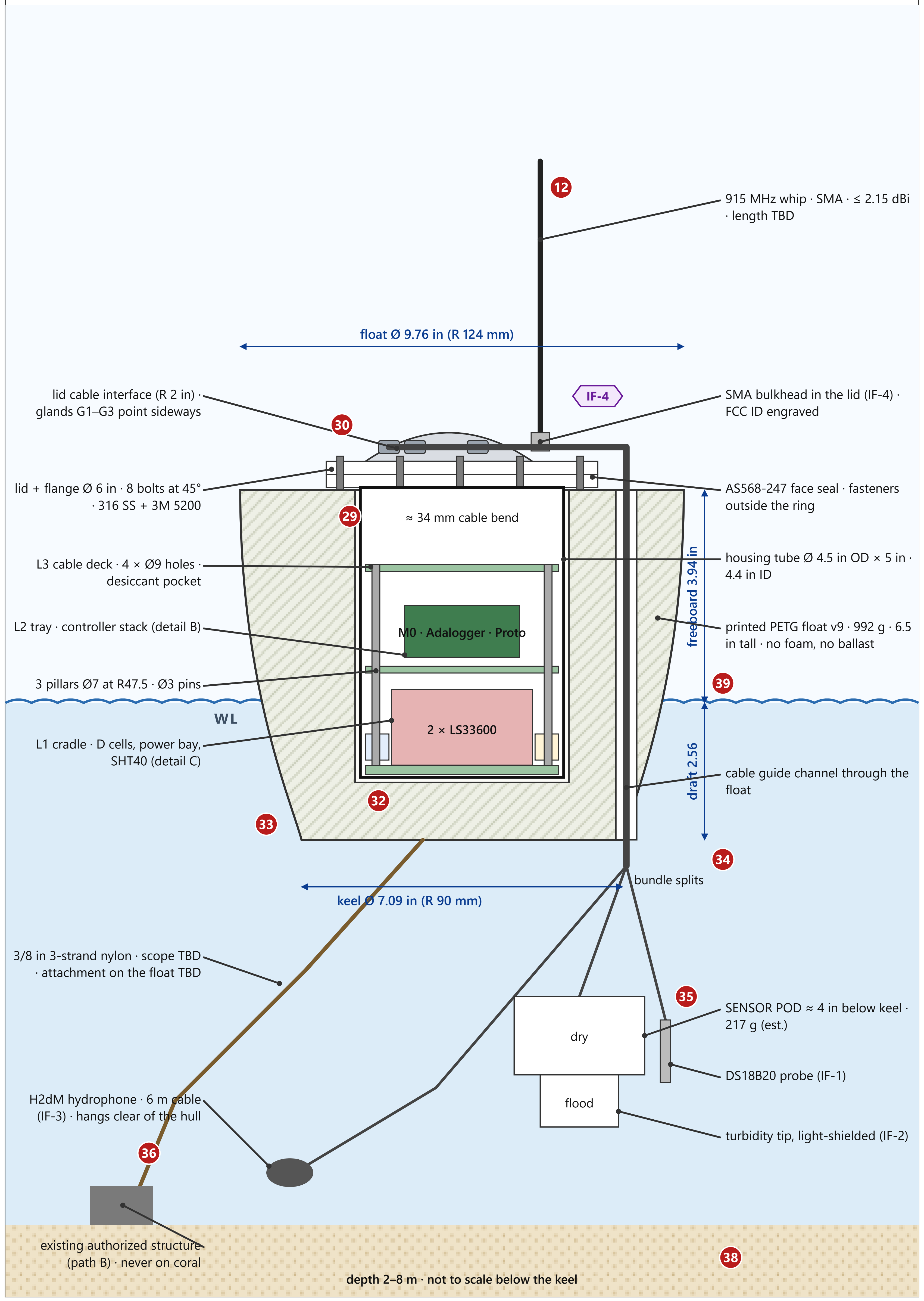
Mechanical and integration

zoom of sheet 1 · sections M, N, P, Q, S, R · same note numbers

PRELIMINARY · NOT FOR FINAL BUILD

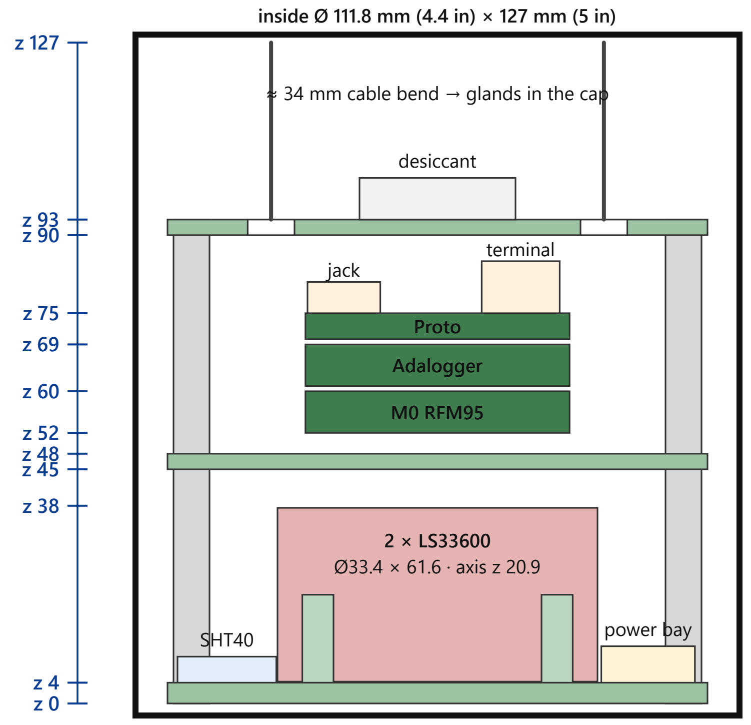
M AS-DEPLOYED BUOY · SECTION

float v9 + flange housing + pod · schematic, NTS


12
915 MHz whip · SMA · ≤ 2.15 dBi · length TBD
float Ø 9.76 in (R 124 mm)
IF-4
lid cable interface (R 2 in) · glands G1–G3 point sideways
30
SMA bulkhead in the lid (IF-4) · FCC ID engraved
lid + flange Ø 6 in · 8 bolts at 45° · 316 SS + 3M 5200
AS568-247 face seal · fasteners outside the ring
29
≈ 34 mm cable bend
L3 cable deck · 4 × Ø9 holes · desiccant pocket
housing tube Ø 4.5 in OD × 5 in · 4.4 in ID
freeboard 3.94 in
L2 tray · controller stack (detail B)
M0 · Adalogger · Proto
printed PETG float v9 · 992 g · 6.5 in tall · no foam, no ballast
3 pillars Ø7 at R47.5 · Ø3 pins
39
WL
2 × LS33600
L1 cradle · D cells, power bay, SHT40 (detail C)
draft 2.56
cable guide channel through the float
32
33
34
bundle splits
keel Ø 7.09 in (R 90 mm)
3/8 in 3-strand nylon · scope TBD · attachment on the float TBD
35
dry
SENSOR POD ≈ 4 in below keel · 217 g (est.)
DS18B20 probe (IF-1)
flood
turbidity tip, light-shielded (IF-2)
H2dM hydrophone · 6 m cable (IF-3) · hangs clear of the hull
36
existing authorized structure (path B) · never on coral
38
depth 2–8 m · not to scale below the keel

N ELECTRONICS HOUSING · PACKAGING OPTION C

side section · z above housing floor, mm


31
inside Ø 111.8 mm (4.4 in) × 127 mm (5 in)
z 127
z 93
z 90
z 75
z 69
z 60
z 52
z 48
z 45
z 38
z 4
z 0
≈ 34 mm cable bend → glands in the cap
desiccant
jack
terminal
Proto
Adalogger
M0 RFM95
2 × LS33600
Ø33.4 × 61.6 · axis z 20.9
SHT40
power bay

93 of 127 mm used · ≈ 34 mm left for cable bend under the cap

L3 deck z 90–93 · 3 mm · 4 × Ø9 cable holes at (±32, ±12) · desiccant pocket 30 × 14 × 8

L2 tray z 45–48 · 4 × Ø6 × 4 standoffs at (±22.9, ±8.9) · boards z 52–75 · jack and terminal on the Proto

pillars Ø7 at R47.5 (90°, 210°, 330°) · 41 mm L1–L2, 38 mm L2–L3 · Ø3 pins or M3 rod in Ø3.2 holes

L1 cradle z 0–38 · 4 mm floor · 2 saddle ribs 69 × 6 × 16.9 with Ø33.8 notches · cells along X at y ±17.2

power bay at x 40.5: diode pockets 15 × 8, PTC 9 × 4, JST 10 × 7 · **SHT40** pocket at x −40.5

㉘ ㉜

LEGEND

- battery 3.6 V
- 3.3 V rail / switched rails
- ground
- 1-Wire data
- analog signal
- I²C
- GPIO on/off control
- RF / LoRa
- data / firmware flow
- planned or physical-retrieval path
- cable (mechanical view)
- mooring line

⓪ note number (see NOTES)

IF-n interface tag (see register S)

† proposed value, confirm on the bench

- 🟩 VERIFIED
- 🟨 UNVERIFIED / OPEN
- 🟥 GAP · NOT BUILT

▨ electrical part	▨ power
▨ firmware	▨ shore station
▨ data / analytics	▨ mechanical

P SEALS · PENETRATIONS · MATERIALS

mechanical/cad READMEs + facts.md

Item	Spec	Status
Housing seal	AS568-247, ID 117.07 × CS 3.53 mm · groove ID 117.07 / OD 122.00, 4.93 wide × 2.65 deep → 25% squeeze, 75% fill · fasteners outside the ring	**confirm ring (SCO-137)**
Flange + lid	Ø6 in flange, 8 bolts at 45° (Ø0.22 / Ø0.13 in holes) · tube Ø4.5 in OD × 5 in · R2 in curved cable interface · FCC ID engraved	**DFM drawing, not released**
Pod seal	Everbilt 2-5/8 OD × 3/16 in CS · groove Ø2.20 × Ø2.74 in, ≈ 28% squeeze · 6 bolts on Ø3.10 in · leads potted in epoxy with an anti-wick boss	**leaked twice; re-test with real ring**
Glands	lab stock, sized per cable · 3 in the cap (+ SMA bulkhead)	**positions not drawn**
Hardware	316 stainless · 3M 5200 over bolt heads and joints	**decided 10-16**
Body + coating	PETG (ASA for scale-up) · Sea Hawk Smart Solution antifouling, copper-free	**decided**
Pressure	5 m water-equivalent (50.3 kPa) before any deployment	**not tested**

㊲

Q MASS · BUOYANCY · STABILITY

printed-float tip explorer, v9 re-run 2026-10-16

Quantity	Value
Total mass (estimate)	2.87 kg
Printed float v9	992 g
Draft / freeboard	2.56 / 3.94 in
KG / GM	2.26 / 1.47 in
Max righting arm	40 mm at ≈ 70°
GZ at 178° (v8)	≈ 1 mm
Float envelope	7.0 L · 6.5 in tall
+2 kg payload	2.1 in freeboard left

Mass inputs (estimates): battery 250 g at 25 mm · boards at 80–90 mm · antenna at 150 mm · epoxy/coating 330 g · mooring hardware 220 g at the keel · pod 217 g + hydrophone 150 g, 4 in below keel. ㉝

No ballast, no lead, no foam. 500 g at the keel only lifts GZ(178°) from 1.1 to 1.5 mm, so hull shape governs. Pod depth moves GM: at the keel 1.20 in, 13.5 in below 1.96 in.

Not covered: mooring heel, wave loads, chamber flooding, fouling mass. Physical tilt test from 180° (SCO-171).

㉞

S INTERFACE REGISTER · HOW THE THREE VIEWS CONNECT

each IF tag appears on the mechanical, electrical and firmware drawings

IF	Mechanical (where it physically is)	Electrical (part → pin)	Firmware / data (what it becomes)	Status
IF-1	DS18B20 probe at the pod · cable up the float channel · gland G1 in the cap	T1 screw terminal → **D12** (1-Wire), R1 4.7 kΩ	temp_c (CSV) · temp_c_centi (packet) · TEMP_TIMEOUT	**parts in hand**
IF-2	Turbidity tip in the pod flood chamber · potted leads · gland G2	Grove board → **A1**; power via Q1 (**D11** LOW)	turbidity_adc (packet) · turbidity_v (SD) · NTU later	**board location open**
IF-3	H2dM on 6 m cable, below the pod · gland G3	J1 jack → preamp → **A2**; power via Q2 (**A3** LOW)	/AUDIO/YYYYMMDD/HHMMSSZ.WAV on SD · audio_present flag · never over LoRa	**not bought · no code**
IF-4	SMA bulkhead in the lid · whip above the waterline	RFM95 uFL → 851 → SMA, ≤ 2.15 dBi, +11 dBm	30 B packet · 3 slots × 3 copies · OTA downlink planned	**regs unverified**
IF-5	2 D cells on their side in the L1 cradle, vents away	diodes → PTC → JST → **BAT**; ÷2 → **A7**	battery_mv → power mode NORMAL / CONSERVE / CRITICAL	**cells not bought**
IF-6	SHT40 in the L1 floor pocket, sealed bay	STEMMA QT → I²C **0x44**	internal_humidity_pct (future column) → leak alarm	**no firmware**
IF-7	microSD in the Adalogger, reached by opening the lid	SPI, CS **D10**	/DATA/YYYYMMDD.CSV, the complete record → sd_import on retrieval	**fixed in code (G4)**
IF-8	Shore station antenna at the shore site (wall power)	ESP32-S3 + RFM95W, erratum applied	decode + dedupe → WiFi push → dashboard (back end open)	**no S3 firmware**

NOTES · same numbers as sheet 1

⑫ **Never power the RFM95 without an antenna.** Deployed antenna ≤ 2.15 dBi and conducted power ≤ 11.6 dBm (FCC ID 2ASEORFM95C, DTS); firmware uses +11 dBm. The housing is engraved “Contains FCC ID: 2ASEORFM95C”.

㉘ SHT40 internal humidity is the leak detector: near-zero RH in a sealed bay, so a rise means a leak. It is on the BOM; firmware and schema columns are not added yet (needs a schema_version bump).

㉙ Electronics housing: 4.4 in ID × 5 in (111.8 × 127 mm), flange style, static face seal on an AS568-247 O-ring (ID 117.07 × CS 3.53 mm) with every fastener outside the ring. Confirm the ring against a supplier table (SCO-137). Never seal an unverified board (Stage 8).

㉚ All cables enter through glands in the top cap: hydrophone, antenna, DS18B20, turbidity. L3 has 4 × Ø9 mm cable holes. Glands come from lab stock; gland positions in the cap are not drawn yet. The 3.5 mm plug (≈ 6 mm) must pass its gland or be fitted after it.

㉛ Packaging option C (2026-10-15): Ø104 mm printed discs on 3 pillars Ø7 at R47.5 (90°/210°/330°) with Ø3 pins or M3 rod. Three layers use 93 of 127 mm, leaving ≈ 34 mm for cable bend. Stack height ≈ 30 mm is unverified, and whether the lid uses part of the 127 mm is not recorded.

㉜ The D cells lie on their side in L1 (Ø33.4 × 61.6 mm, ≈ 90–100 g each), vents facing away from the electronics.

㉝ Float v9 (2026-10-16): printed PETG, 992 g, no foam, no ballast stem, no lead, with a cable guide channel through the float. At 2.87 kg: draft 2.56 in, freeboard 3.94 in, GM 1.47 in; self-rights from every angle, but GZ is only ≈ 1 mm near 178°. Tilt test from 180° pending (SCO-171).

㉞ Stability masses are estimates (battery 250 g, epoxy/coating 330 g, mooring hardware 220 g, pod 217 g, hydrophone 150 g). Replace them with weighed values and re-run run_printed_float.py.

㉟ Sensor pod hangs ≈ 4 in below the keel (study value). The dry chamber holds electronics; the flood chamber shades the probe from light. Pod O-ring: Everbilt 2-5/8 in OD × 3/16 in CS. Turbidity leads are epoxy-potted, not O-ring sealed.

㊱ Mooring: tether to an existing authorized structure (permitting path B), no anchor of our own. Line: 3/8 in 3-strand nylon, scope TBD (SCO-17). Attachment hardware on the printed float is undecided. Fallback at an unmarked site: one mushroom anchor beside, never on, coral.

㊲ Corrosion: 316 stainless hardware with 3M 5200 over exposed bolt heads and joints. Antifouling: Sea Hawk Smart Solution (copper-free; copper is toxic to coral). Fouling drift is screened in analytics (drift.py).

㊳ Deployment depth 2–8 m. Pressure-test the sealed housing to 5 m water-equivalent before deployment. Survival design set: d 2 m · H 1.2 m · T 6 s · current 0.8 m/s · wind 22 m/s.

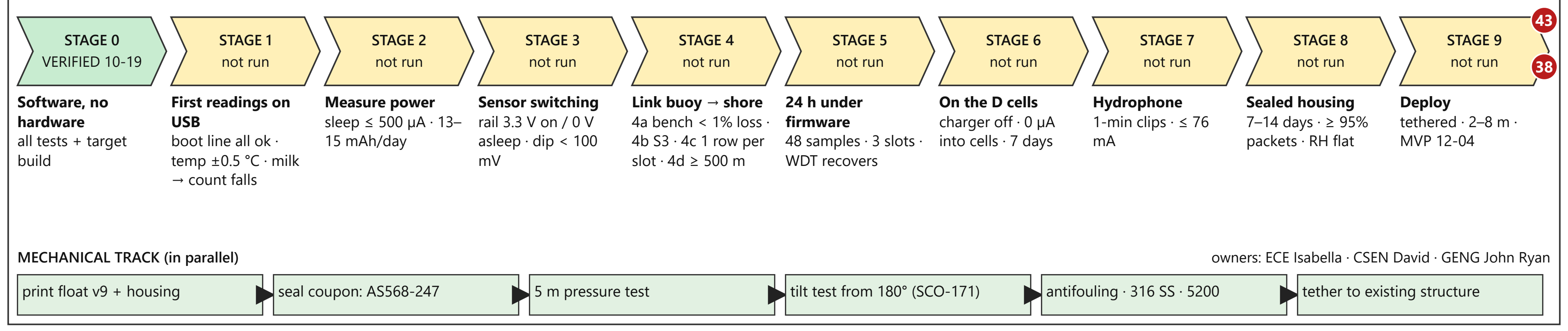
㊴ Handling: set the buoy down on its base, never on a sidewall.

㊸ MVP target 2026-12-04 (hard limit 12-11): the buoy floats and self-rights, takes one temperature and one turbidity reading, and a packet reaches shore.

㊻ Sources: repo main @ 455f709 (2026-10-19, through PR #187: runbook rewrite, bench readiness, agreed shore design); Rev B drawing set SCOUT-E-100 (2026-10-12); power budget update 2026-10-05; packaging option C (2026-10-15); printed-float v9 study (2026-10-16); facts.md.

R BUILD, TEST + DEPLOY WORKFLOW

build-and-test plan stages · each stage's PASS is the next stage's precondition


STAGE 0
VERIFIED 10-19
STAGE 1
not run
STAGE 2
not run
STAGE 3
not run
STAGE 4
not run
STAGE 5
not run
STAGE 6
not run
STAGE 7
not run
STAGE 8
not run
STAGE 9
not run
43
38

Software, no hardware
all tests + target build

First readings on USB
boot line all ok · temp ±0.5 °C · milk → count falls

Measure power
sleep ≤ 500 µA · 13–15 mAh/day

Sensor switching
rail 3.3 V on / 0 V asleep · dip < 100 mV

Link buoy → shore
4a bench < 1% loss · 4b S3 · 4c 1 row per slot · 4d ≥ 500 m

24 h under firmware
48 samples · 3 slots · WDT recovers

On the D cells
charger off · 0 µA into cells · 7 days

Hydrophone
1-min clips · ≤ 76 mA

Sealed housing
7–14 days · ≥ 95% packets · RH flat

Deploy
tethered · 2–8 m · MVP 12-04

MECHANICAL TRACK (in parallel)

owners: ECE Isabella · CSEN David · GENG John Ryan

print float v9 + housing → seal coupon: AS568-247 → 5 m pressure test → tilt test from 180° (SCO-171) → antifouling · 316 SS · 5200 → tether to existing structure

PROJECT
S.C.O.U.T. · Santa Clara Oceanic Utilities Transmitter
Nearshore reef monitoring buoy · Santa Clara University Senior Design 2026–27

TITLE
Mechanical and integration

DRAWING NO. **SCOUT-S-000**	SHEET **4 of 4**
REVISION **A (draft) · first issue**	DATE **2026-10-19**
DRAWN **I. Rodriguez (ECEN)**	CHECKED **— (S. Krishnan to review)**
MECH SOURCE J. R. Myrdal (GENG) CAD	FW / SHORE SOURCE D. Chousal (CSEN) repo
PAPER · SCALE 22 × 17 in (ANSI C) · NTS	SUPERSEDES nothing · companion to SCOUT-E-100 Rev B